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Cross-layer contagion of prompt injections in multi-agent swarms: a multiplex microscopic markov chain approach

Tran Duc Le¹ , Truong Duy Dinh^{2*}, Thi Le Quyen Nguyen³ and Cong Danh Nguyen³

Abstract

Large Language Models (LLMs) are increasingly deployed as interconnected agentic swarms that leverage the Model Context Protocol (MCP) to invoke shared external tools, APIs, and databases. In these settings, conventional security strategies based on agent-to-agent airgapping can be insufficient because agents that never directly communicate may still cross-infect one another through shared infrastructure tools, producing a “Confused Deputy” cascade. This paper develops a formal mathematical framework for modeling this cross-layer contagion in MCP-enabled multi-agent swarms. We construct a *coupled multiplex Microscopic Markov Chain Approach* (MMCA) that simultaneously tracks the node-level probability flow across two layers: an Agent cognitive layer (governed by Susceptible–Exposed–Infected–Quarantined (SEIQ) dynamics) and a Tool infrastructure layer (governed by Susceptible–Infected–Susceptible (SIS) dynamics). Our contributions are threefold: (i) we formulate a coupled multiplex MMCA with asymmetric SEIQ–SIS dynamics across agent and tool layers; (ii) we derive an analytical characterization of systemic risk, including an epidemic-threshold approximation via Next-Generation Matrix analysis and a closed-form budget-allocation rule under an exponential defense-efficiency model; and (iii) across eight experiment groups on synthetic and empirical agent-layer topologies, we show that shared-tool coupling consistently amplifies contagion and that tool-side controls can dominate agent-side hardening under the modeled regime. For cybersecurity practice, the framework identifies when shared tool infrastructure can transform localized prompt-injection events into system-level risk even under direct agent-to-agent isolation.

Keywords Multi-agent systems, Prompt injection, Epidemic modeling, Multiplex networks, Microscopic markov chain approach, MCP security, Network contagion

Introduction

The architecture of applications powered by Large Language Models (LLMs) is shifting from single-agent chat interfaces toward *multi-agent swarms* - collaborative

networks of specialized LLMs that autonomously decompose tasks, delegate sub-routines, and synthesize results (Wu et al. 2026; Talebirad and Nadiri 2023; Hong et al. 2024). Frameworks such as AutoGen (Wu et al. 2026), LangChain (Chase and Langchain 2022), CrewAI, and MetaGPT (Hong et al. 2024) have enabled increasingly complex orchestrated workflows in software engineering, financial analysis, scientific discovery, and autonomous research.

A central enabler of this architecture is the *Model Context Protocol* (MCP) (Gullí 2025), a standardized interface allowing agents to dynamically connect to external context sources, execution environments,

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vector databases, and code interpreters. By abstracting tool invocation behind a unified protocol, MCP enables agents to access external systems and mutate shared state. However, intertwining multi-agent networks with external tool execution introduces an attack surface that escapes traditional cybersecurity paradigms.

One relevant threat in this landscape is the *indirect prompt injection* attack (Greshake et al. 2023; Zhan et al. 2024; Liu et al. 2024; Yi et al. 2025), wherein malicious instructions hidden in seemingly benign external documents - webpages, resumes, database records - hijack the reasoning engine of an LLM upon ingestion. In a single-agent system, the blast radius is typically confined to the hijacked session. In an interconnected swarm, that blast radius can extend beyond the initial compromise through shared tools and downstream agent interactions. To mitigate cascading failures, system architects frequently rely on “airgapping” strategies-enforcing strict access controls or zero-trust cryptographic policies on agent-to-agent communication channels (Ruan et al. 2023). The prevailing assumption is that prohibiting direct inter-agent messaging prevents cross-infection.

This assumption overlooks the underlying *topological reality* of tool-augmented swarms. Agents do not merely exist in a flat communication graph; they operate across a *multiplex network* (Gómez et al. 2013; De Domenico et al. 2013; Boccaletti et al. 2014) where they simultaneously reside in an agent collaborative layer and a shared tool-access layer. When multiple agents access a shared MCP tool-a common vector database, a shared code interpreter, or a global memory scratchpad-the tool acts as an implicit, hidden bridge. This manifests the classic “Confused Deputy” problem (Hardy 1988) at a swarm level. For example, Agent A (a web researcher) is compromised by a malicious website and poisons a shared SQL database. Agent B (a trusted finance orchestrator), which is entirely airgapped from Agent A, subsequently queries the database, ingests the payload, and executes an unauthorized transaction. Conventional agent-to-agent firewalling is bypassed entirely, allowing the contagion to pivot across the tool layer and amplify back into the trusted swarm.

To formalize and mathematically quantify this vulnerability, we propose a *Microscopic Markov Chain Approach* (MMCA) (Gómez et al. 2010; Wang et al. 2017) specifically adapted for cross-layer contagion in multi-agent swarms. Originating from complex network science to model biological epidemics and information cascades, MMCA offers exact node-level topological tracing of state transitions. We extend this foundation to a bipartite, two-layer architecture, explicitly modeling non-linear feedback loops between Agent nodes

(following Susceptible–Exposed–Infected–Quarantined (SEIQ) dynamics) and Tool nodes (following Susceptible–Infected–Susceptible (SIS) dynamics).

Our contributions are as follows:

- *Multiplex MMCA formulation*: We present a discrete-time MMCA model that simultaneously captures SEIQ agent dynamics and SIS tool dynamics across a two-layer multiplex network, with three independent transmission pathways ($\beta_A, \beta_B, \beta_T$).
- *Analytical risk characterization*: We derive an analytical approximation for the epidemic threshold R_0 under annealed mean-field assumptions via Next-Generation Matrix (NGM) analysis, and under an exponential defense-efficiency model we obtain a closed-form budget-allocation rule showing how the optimal split between agent-side and tool-side defenses follows an inverse-efficiency law.
- *Systematic empirical characterization*: Across 8 experiment groups spanning Monte Carlo alignment (Mean Absolute Error (MAE) ≤ 0.012), phase-transition verification, cross-layer amplification analysis, six structural ablations plus a topology comparison, sensitivity studies, architectural scenario benchmarks, and three empirical agent-layer datasets with synthetic tool layers, we show that shared-tool coupling consistently amplifies contagion and that tool-side controls can dominate agent-side hardening under the modeled regime.

The remainder of this paper is organized as follows. Section [Related work](#) reviews related work. Section [The cross-layer contagion framework](#) presents the mathematical framework. Section 3 derives the epidemic threshold. Section [Security resource optimization](#) formulates the security budget optimization. Section [Experimental validations](#) reports experimental results. Section [Discussion](#) discusses implications and limitations, and Section [Conclusion](#) concludes.

Related work

Our work sits at the intersection of three research streams: (i) mathematical epidemic modeling on networks, (ii) multiplex network dynamics, and (iii) LLM agent security. We additionally review multi-agent frameworks and security economics, which inform our system model and optimization formulation, respectively.

Cyber-epidemic modeling

The application of epidemiological models to computer security traces back to the seminal work of Kephart and White (Kephart and White 1991), who adapted the SIS compartmental model to analyze computer virus

propagation on directed graphs. Their key insight - that the epidemic threshold depends on the network's spectral properties - laid the groundwork for an entire subfield of network-based security modeling.

Pastor-Satorras and Vespignani (2001) proved that in scale-free networks (where the degree distribution follows a power law), the epidemic threshold vanishes as the network size grows. This result had profound implications for malware defense, since many real-world computer networks exhibit heavy-tailed degree distributions. Their subsequent work on targeted immunization (Pastor-Satorras and Vespignani 2002) demonstrated that degree-based vaccination strategies are exponentially more effective than random immunization in heterogeneous networks. Van Mieghem et al. (2009) advanced the field by introducing the *N*-Intertwined Mean-Field Approximation (NIMFA), which computes exact per-node infection probabilities while remaining computationally tractable for large networks. Chakrabarti et al. (2008) derived tight eigenvalue-based epidemic thresholds and showed that the largest eigenvalue of the adjacency matrix governs the phase transition between disease-free and endemic equilibria.

More recent surveys by Nowzari et al. (2016) and Paré and Beck (2020) have systematically cataloged epidemic processes on networks with applications to cybersecurity, including botnet propagation, worm spreading, and information cascades. Xu et al. (2012) extended epidemic models to incorporate prophylactic and reactive countermeasures, providing game-theoretic foundations for optimal patch deployment. Despite this rich literature, existing models treat all network nodes as homogeneous computing devices residing in a *single* topological layer. This assumption does not hold in multi-agent LLM swarms, where agents and tools occupy functionally distinct roles with asymmetric state dynamics - a gap that motivates our multiplex formulation.

Microscopic markov chain approach on multiplex networks

Classical epidemic models - both mean-field (Pastor-Satorras and Vespignani 2001) and heterogeneous mean-field (Van Mieghem et al. 2009) - operate at a macroscopic level, tracking aggregate fractions of infected nodes rather than individual node states. (a) Gómez et al. (2010) introduced the MMCA to overcome this limitation, deriving exact node-level transition probabilities that preserve the full topological heterogeneity of the contact network. In MMCA, each node i maintains an independent probability vector over its compartmental states, and the per-step update equations encode the precise neighborhood structure through product-form non-infection probabilities. This yields a deterministic

dynamical system that provably converges to the same steady state as the stochastic process in the limit of large populations.

The framework was subsequently extended to *multiplex networks* - networks with multiple interaction layers connecting the same set of nodes but through different types of relationships. (b) Gómez et al. (2013) analyzed diffusion dynamics on interconnected layers, proving that inter-layer coupling can dramatically alter the diffusion time and spectral properties relative to isolated layers. De Domenico et al. (2013) provided the foundational mathematical formalism for multilayer networks using tensorial representations, enabling rigorous spectral analysis of layer-coupled dynamics. Granell et al. (2013) demonstrated a particularly relevant phenomenon: the interplay between an awareness layer and a disease-spreading layer on the same multiplex produces *metacritical* phase transitions that are qualitatively absent in single-layer models. Cozzo et al. (2013) showed that even weak inter-layer coupling can push the epidemic threshold below that of any individual layer - a precursor to the cross-layer amplification we observe in our agent-tool system.

Boccaletti et al. (2014) provided a comprehensive review of multilayer network structure and dynamics, covering spectral theory, percolation transitions, and synchronization phenomena. Wang et al. (2017) unified several competing mean-field theories (NIMFA, MMCA, pair approximation) and characterized their relative accuracy. Sanz et al. (2014) specifically studied SIS-SIS multiplex epidemics, demonstrating the existence of an "epidemic invasion threshold" where a disease endemic in one layer can invade a previously disease-free layer.

Our work adapts this multiplex MMCA machinery to the domain of AI systems security through an explicit agent-tool contagion model. We depart from prior multiplex epidemic studies in two ways: (i) the two layers have *distinct* node types (agents vs. tools, not the same nodes in different contexts), producing a bipartite cross-layer coupling rather than a node-aligned multiplex; and (ii) the compartmental dynamics are *asymmetric* across layers (SEIQ for agents, SIS for tools), reflecting the differing operational semantics of cognitive agents and stateless infrastructure.

LLM agent security and prompt injection

The security of LLM-based applications has emerged as an active research area. Schulhoff et al. (2023) systematically studied prompt injection as a vulnerability class through large-scale competition, demonstrating that adversarial instructions can reliably override system-level guardrails. Greshake et al. (2023) elevated this threat to *indirect prompt injection*, where malicious payloads are

not supplied by the user directly but are instead hidden in external documents - webpages, emails, or database records - that the LLM retrieves and processes as part of its augmented workflow. This attack modality bypasses user-facing input filters entirely.

Subsequent work has expanded understanding along several dimensions. Zhan et al. (2024) introduced INJEAGENT, a comprehensive benchmark for evaluating the vulnerability of tool-integrated LLM agents to indirect prompt injections across 17 different tool categories. Liu et al. (2024) formalized prompt injection taxonomies, distinguishing between goal hijacking, prompt leaking, and denial-of-service attacks. Yi et al. (2025) developed systematic benchmarks for evaluating agent robustness under various attack strategies. Cohen et al. (2024) demonstrated “here comes the AI worm” - a self-replicating prompt injection that can propagate through GenAI ecosystems by embedding itself in model outputs that become inputs for downstream agents, establishing the first empirical evidence of worm-like contagion in multi-agent systems. On the defense side, several mitigation strategies have been proposed, including instruction hierarchy enforcement (Wallace et al. 2024), prompt boundary demarcation, and output sanitization. However, Ruan et al. (2023) demonstrated through extensive red-teaming that multi-agent systems exhibit emergent vulnerabilities not present in single-agent deployments, including privilege escalation through inter-agent trust and cascading failures through shared memory. Gu et al. (2024) surveyed agent-specific safety concerns, concluding that the transition from isolated LLMs to interconnected agent systems creates qualitatively new threat surfaces.

More recent work has begun to study the same problem from protocol and enforcement perspectives: Costa et al. (2025) use information-flow control (IFC) to constrain agent behavior, Ferrag et al. (2026) survey prompt-to-protocol exploits across agent workflows, Jamshidi et al. (2025) analyze MCP-specific tool-poisoning and adversarial attack surfaces, and Palumbo et al. (2026) compile secure policies into agentic systems.

Crucially, our epidemic-modeling approach provides a macro-level, probabilistic complement to deterministic, micro-level enforcement mechanisms like IFC. While IFC tracks explicit taint propagation across specific agent-tool interactions to block unauthorized flows at runtime, it can struggle with the semantic ambiguity of natural language and the massive state space of interconnected swarms. In contrast, our framework does not require tracking individual execution traces; instead, it models the systemic vulnerability of the architecture itself, allowing architects to proactively tune the network topology and allocate defense

budgets to minimize steady-state contagion risk. Both approaches are necessary: IFC provides local runtime enforcement, whereas our epidemic model guides global architectural resilience. The security of multi-agent systems is an actively evolving domain, and securing shared infrastructure like MCP has become an important operational concern for enterprise deployments.

A recurring limitation in the existing security literature is its treatment of prompt injection primarily as a *local*, agent-level phenomenon: a specific agent is compromised, and the analysis focuses on the blast radius of that individual compromise. To the best of our knowledge, prior work has not mathematically modeled the *systemic, network-level contagion dynamics* of prompt injections propagating across an interconnected swarm of heterogeneous agents operating through shared infrastructure - the specific gap our framework addresses.

Multi-agent frameworks and tool integration

The practical deployment of multi-agent LLM systems has been catalyzed by a proliferation of orchestration frameworks. AutoGen (Wu et al. 2026) introduced the paradigm of multi-agent conversation, enabling flexible topologies where specialized agents collaborate through structured dialogue with Human-in-the-Loop (HITL) capability. LangChain (Chase and Langchain 2022) popularized the concept of “chains” and “agents” that can dynamically invoke external tools. MetaGPT (Hong et al. 2024) demonstrated that encoding standardized operating procedures into multi-agent workflows significantly improves collaborative code generation quality. Talebirad and Nadiri (2023) provided a taxonomy of multi-agent collaboration patterns, distinguishing between cooperative, competitive, and hierarchical architectures.

The MCP (Gullí 2025) represents a significant standardization effort, providing a unified interface for LLM agents to discover, connect to, and invoke external tools - ranging from code interpreters and web browsers to database clients and file systems. By abstracting tool invocation behind a standardized protocol, MCP enables agents to dynamically compose capabilities at runtime. However, this very composability creates a shared-state surface that can be exploited for cross-agent contagion: when two agents access the same MCP tool, they implicitly share a communication channel that bypasses any agent-to-agent access controls.

These frameworks reveal an inherent architectural tension: the more flexibly agents can share tools and resources, the larger the attack surface for cross-layer contagion. Our framework offers a theory-focused mathematical characterization of this trade-off through the lens of multiplex epidemic theory.

Security investment optimization

The economics of information security investment has been studied since the seminal Gordon–Loeb model (Gordon and Loeb 2002), which proved that the optimal level of investment in information security is always less than the expected loss from a breach, and that investment should be concentrated on assets with intermediate vulnerability levels rather than the most or least vulnerable ones. Subsequent extensions incorporated dynamic investment strategies (Cavusoglu et al. 2008), network externalities in interconnected systems (Kunreuther and Heal 2003), and game-theoretic interactions between attackers and defenders (Laszka et al. 2015).

In the context of network epidemics, optimal resource allocation for containing spreading processes has been studied by Drakopoulos et al. (2017) and Preciado et al. (2014), who formulated convex optimization problems for vaccine allocation and contact rate reduction. Our optimization formulation (Section Security resource optimization) builds on this tradition but introduces a novel dimension: the allocation is not between different nodes within a single network but between *different layers* of a multiplex architecture, leading to the counterintuitive inverse-efficiency allocation law of Theorem 4.

Positioning and differentiation

Table 1 summarizes the positioning of our work relative to the closest prior studies across five key dimensions. Relative to this author-defined comparison class, our framework is the only one in Table 1 that combines all five dimensions - multiplex topology, microscopic (node-level) resolution, asymmetric cross-layer dynamics, closed-form epidemic threshold, and budget optimization - within a single AI-security formulation. The table should be interpreted as an author-side synthesis of the closest comparison class rather than as an exhaustive benchmark of all adjacent literature.

The cross-layer contagion framework

Problem statement

The core problem addressed in this paper is the formal quantification of vulnerability in multiplex AI systems, where seemingly secure agents can be compromised via shared infrastructure. When an LLM swarm is deployed, security architects typically enforce boundaries by restricting direct agent-to-agent communication (layer A). However, these agents must interact with a shared set of MCP tools (layer B) to perform useful work. If an external attacker injects a malicious payload into a tool, the tool becomes a passive vector for continuous infection. The challenge is that traditional models evaluate contagion on single-layer networks, completely missing the *cross-layer feedback loop* where an agent poisons a tool, and that tool subsequently infects an arbitrary number of other airgapped agents, as illustrated in Fig. 1. Consequently, it is necessary to construct a unified mathematical framework that simultaneously evaluates the coupled SEIQ (agent) and SIS (tool) dynamics to derive the true system-wide epidemic threshold (R_0) and optimize security budget allocations.

In this section, we develop the mathematical framework that captures cross-layer prompt injection contagion in multi-agent LLM swarms. The central modeling challenge is that agents and tools play fundamentally different roles in the infection cycle: agents are cognitive entities that can be compromised, enter a latent

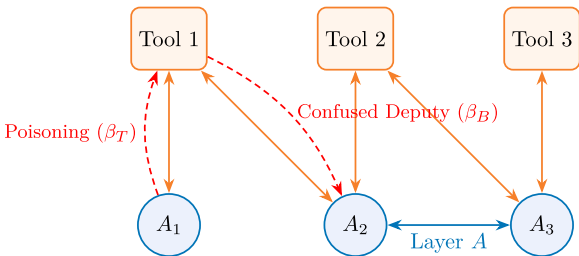


Fig. 1 Conceptual Visualization of Cross-Layer Contagion. Agents interact directly (Layer A) and indirectly via shared MCP tools (Layer B). A poisoned tool acts as a contagion bridge, pivoting infections between airgapped agents ($A_1 \rightarrow T_1 \rightarrow A_2$)

Table 1 Positioning relative to prior work. ✓ = fully addressed; o = partially addressed; - = not addressed

Study	Multiplex	Microscopic	Cross-Layer	R_0 Threshold	Budget Opt.
Kephart and White (1991)	-	-	-	o	-
Pastor-Satorras and Vespignani (2001)	-	-	-	✓	-
Van Mieghem et al. (2009)	-	✓	-	✓	-
(a) Gómez et al. (2010)	-	✓	-	✓	-
(b) Gómez et al. (2013)	✓	o	o	-	-
Granell et al. (2013)	✓	✓	o	✓	-
Cohen et al. (2024)	-	-	o	-	-
This Paper	✓	✓	✓	✓	✓

incubation period, exhibit active malicious behavior, and eventually be detected and quarantined; tools, by contrast, are stateless infrastructure components that are either clean or poisoned, with no notion of incubation or quarantine - they simply serve contaminated content until patched. This asymmetry dictates our choice of compartmental models: a four-state SEIQ process for agents, and a simpler two-state SIS process for tools. The two layers are coupled through the bipartite agent-tool interaction matrix, creating a feedback loop where infected agents can poison tools, and poisoned tools can, in turn, infect other agents.

We choose SEIQ for the agent layer rather than simpler alternatives such as SIS, SIR, or SIQ for three reasons. First, an *exposed* state is needed because prompt-injection payloads need not trigger immediately after ingestion; an agent may carry a malicious instruction in context and only activate it in a later reasoning step. Second, a *quarantined* state is needed because operational defenses are typically isolation-based: suspicious agents are disabled, sandboxed, or stripped of tool permissions before reintroduction. Third, quarantined agents return to the susceptible state because deployed LLM agents are usually reset and returned to service rather than permanently removed, so reinfection remains possible. Thus, SEIQ is the smallest model here that captures delayed activation, active propagation, isolation, and return to service.

We note that this framework captures *topological co-exposure pathways* - the structural possibility of cross-infection through shared tool access - rather than the semantic details of how a specific prompt injection payload exploits a specific LLM's instruction-following behavior. The scalar transmission probabilities ($\beta_A, \beta_B, \beta_T$) abstract over the complex chain of events by which a malicious payload enters an agent's context window, survives filtering, and influences output; this abstraction enables tractable mathematical analysis at the cost of discarding token-level attack semantics.

System model and threat assumptions

We model an MCP-enabled AI swarm as a *multiplex network* consisting of two layers operating over a shared set of N LLM agents and M MCP tools. The term "multiplex" reflects the fact that the same set of agents participates simultaneously in two distinct interaction layers - an *agent cognitive layer* (Layer A) governing direct inter-agent communication via dialogue, delegation, and result sharing, and a *tool infrastructure layer* (Layer B) governing the read/write interactions between agents and shared MCP tools. This dual-layer representation is critical because it captures interaction pathways that are invisible to single-layer models: two agents that never

communicate directly can still cross-infect one another by reading from and writing to a common tool.

Definition 1 (*Multiplex Swarm Network*) The swarm is a tuple $\mathcal{G} = (V_A, V_T, A, B)$ where:

- $V_A = \{1, \dots, N\}$ is the set of LLM agents.
- $V_T = \{1, \dots, M\}$ is the set of MCP tools/servers.
- $A = [a_{ij}] \in \{0, 1\}^{N \times N}$ is the agent-to-agent adjacency matrix (**Layer A**).
- $B = [b_{ik}] \in \{0, 1\}^{N \times M}$ is the agent-to-tool adjacency matrix (**Layer B**).

The matrix A encodes the cognitive communication topology: $a_{ij} = 1$ indicates that agent j can send messages (dialogue turns, delegated subtask results, or shared reasoning chains) to agent i . In practice, this corresponds to the orchestration graph defined by frameworks such as AutoGen or LangChain - for example, in a hub-spoke AutoGen deployment, the central orchestrator agent has incoming edges from all specialist agents. The matrix B encodes the tool-access topology: $b_{ik} = 1$ indicates that agent i has read and/or write access to MCP tool k . (*Assumption: We treat tool access as a symmetric read/write relationship. Directed access control mechanisms distinguishing read-only from write access are reserved for future work*). Accordingly, B should be interpreted as a coarse-grained *exposure matrix* that merges read and write permissions into a single effective relation. This matrix is typically determined by the tool registration and permission policies of the MCP server configuration.

Crucially, the product BB^T defines the *implicit agent-to-agent coupling through shared tools*: the entry $(BB^T)_{ij}$ counts the number of tools jointly accessed by agents i and j . Even when $a_{ij} = 0$ (no direct communication), a nonzero $(BB^T)_{ij}$ creates a hidden contagion pathway through the shared tool layer. This observation motivates our multiplex formulation: the matrix BB^T captures precisely the cross-layer bridge that single-layer models cannot represent.

Attacker Model. We consider an external adversary who cannot directly compromise the swarm's internal infrastructure (e.g., API keys, model weights, or system prompts stored server-side), but who can manipulate the *content* of external data sources that MCP tools ingest. Concretely, the attacker may plant a malicious prompt injection payload on a webpage that a RAG-connected retrieval tool will index, inject adversarial instructions into a shared Google Drive document accessed by a file-reading tool, or embed hidden directives in a database record returned by a SQL query tool. This threat model

aligns with the indirect prompt injection attack vector formalized by Greshake et al. (2023).

Once the initial poisoning succeeds, the contagion can propagate through three distinct transmission channels, each governed by its own rate parameter:

- **Tool→Agent transmission (β_B):** A susceptible agent queries a poisoned tool and ingests the malicious payload into its context window. The rate β_B captures the probability that the payload successfully hijacks the agent's reasoning - a function of the agent's input sanitization, prompt boundary enforcement, and instruction hierarchy compliance (Wallace et al. 2024).
- **Agent → Agent transmission (β_A):** An infected agent forwards the malicious payload (or a payload derivative, such as a subtly modified chain-of-thought) to a neighboring agent via dialogue. The rate β_A is typically lower than β_B because agent-to-agent messages often undergo format checking and are less likely to contain raw executable instructions compared to tool responses.
- **Agent → Tool transmission (β_T):** An infected agent writes contaminated content back to a shared tool - for example, updating a vector database with poisoned embeddings, committing malicious code to a shared repository, or inserting adversarial entries into a memory scratchpad. The rate β_T depends on whether write operations are subject to Human-in-the-Loop (HITL) approval or automated sandboxing.

These three channels create a closed feedback loop: Tool→Agent→Agent→Tool→..., where each cycle can amplify the total infection beyond what either layer could sustain in isolation. This cross-layer feedback is the central phenomenon our framework is designed to capture and quantify.

Notation and parameter definitions

Table 2 summarizes the complete notation used throughout this paper.

Several parameter relationships merit discussion. The three transmission rates β_A , β_B , and β_T govern qualitatively different interaction types and are expected to differ in magnitude in practice. In the illustrative regime studied here, the tool-to-agent rate β_B is often expected to be the largest, because tool responses can bypass some of the conversational guardrails that filter agent-to-agent messages - for example, a RAG tool may return retrieved content that the agent treats as trusted context. The agent-to-agent rate β_A is modulated by whatever input sanitization is applied to inter-agent messages (e.g., output format enforcement, content moderation layers).

Table 2 Complete notation and parameter definitions for the Cross-Layer MMCA framework

Symbol	Definition	Domain
N	Number of LLM agents	$\mathbb{Z}^+$
M	Number of MCP tools/servers	$\mathbb{Z}^+$
$A = [a_{ij}]$	Agent-to-Agent adjacency matrix (Layer A)	$\{0, 1\}^{N \times N}$
$B = [b_{ik}]$	Agent-to-Tool adjacency matrix (Layer B)	$\{0, 1\}^{N \times M}$
$p_{i,S}(t)$	Probability agent i is Susceptible at t	$[0, 1]$
$p_{i,E}(t)$	Probability agent i is Exposed at t	$[0, 1]$
$p_{i,I}(t)$	Probability agent i is Infected at t	$[0, 1]$
$p_{i,Q}(t)$	Probability agent i is Quarantined at t	$[0, 1]$
$p_{k,S}^T(t)$	Probability tool k is Susceptible at t	$[0, 1]$
$p_{k,I}^T(t)$	Probability tool k is Infected at t	$[0, 1]$
β_A	Agent→Agent infection rate	$[0, 1]$
β_B	Tool→Agent infection rate (poisoned read)	$[0, 1]$
β_T	Agent→Tool infection rate (malicious write)	$[0, 1]$
σ	Incubation-to-infection activation rate	$[0, 1]$
γ_A	Agent detection/quarantine rate	$(0, 1]$
γ_T	Tool patch/cleaning rate	$(0, 1]$
δ	Agent reboot rate (Quarantined → Susceptible)	$[0, 1]$

The agent-to-tool rate β_T depends heavily on whether tool write operations require HITL approval or are fully automated - a critical architectural choice for system designers.

The incubation rate σ models the delay between initial compromise and active malicious behavior. In LLM agent systems, this corresponds to the latency between when a prompt injection payload enters the agent's context window and when the agent acts on the injected instructions. This delay can range from immediate (if the injected instruction is immediately relevant to the current task) to indefinitely deferred (if the payload is a "sleeper" instruction triggered only by specific future queries). The detection rate γ_A and patching rate γ_T represent the defensive capabilities of the operational environment: γ_A captures behavioral anomaly detection, output auditing, and automated kill-switches on agents, while γ_T captures periodic integrity scans, hash verification, and rollback mechanisms on tools.

Microscopic transition equations

We adopt the *discrete-time MMCA* (Gómez et al. 2010), a stochastic modeling technique that tracks the infection probability of every individual node in the network, as opposed to classical mean-field approaches that track only aggregate population fractions. The key advantage of MMCA for our setting is that it preserves the *heterogeneity* of the multiplex topology: agents with many tool connections or high centrality in the

communication graph will naturally accumulate higher infection probabilities than peripheral agents, enabling the framework to capture the structural vulnerabilities that arise from specific architectural choices (e.g., hub-spoke vs. peer-to-peer agent orchestration).

At each discrete time step $t \rightarrow t + 1$, every node independently transitions between compartmental states according to the infection probabilities dictated by its neighborhood in both layers. In the context of LLM swarms, a single time step abstractly represents a synchronous operational cycle - for example, a reasoning step, a round of delegated subtask generation, or a state-evaluation across all active components. The transition logic proceeds in two stages: first, we compute the probability that each node *survives* all potential infection attempts from its neighbors (the “non-infection probability”); second, we use these survival probabilities to update the full compartmental state vector.

Non-infection probabilities

The central quantity in the MMCA formulation is the *non-infection probability*: the probability that a susceptible node avoids infection from all of its neighbors during a single time step. This probability takes a product form under the MMCA independence assumption - namely, that infection attempts from distinct neighbors are conditionally independent given the current state vector. While this assumption introduces a controlled approximation (it ignores higher-order correlations between simultaneous infections from multiple sources), it is standard in the MMCA literature (Gómez et al. 2010; Van Mieghem et al. 2009) and yields excellent agreement with stochastic Monte Carlo simulations, as we verify empirically in Section [Experimental validations](#).

Agent-layer survival. The probability that a susceptible agent i escapes infection from all of its neighbors in the agent cognitive layer (Layer A) at step t is:

$$q_i^A(t) = \prod_{j=1}^N (1 - \beta_A a_{ji} p_{j,I}(t)) \quad (1)$$

Each factor $(1 - \beta_A a_{ji} p_{j,I}(t))$ represents the probability of surviving a single potential infection attempt from neighbor j : the term a_{ji} ensures that only actual neighbors contribute (non-neighbors have $a_{ji} = 0$), the term $p_{j,I}(t)$ weights the attempt by the probability that j is actually infectious at time t , and β_A is the per-contact transmission probability. Intuitively, an agent with many infected neighbors in Layer A will have a small q_i^A - i.e., a high probability of becoming infected through dialogue.

Tool-layer survival. Analogously, the probability that agent i is *not* infected by reading from any of its connected tools in Layer B is:

$$q_i^B(t) = \prod_{k=1}^M (1 - \beta_B b_{ik} p_{k,I}^T(t)) \quad (2)$$

Here, b_{ik} indicates whether agent i has access to tool k , and $p_{k,I}^T(t)$ is the probability that tool k is currently poisoned. Agents that access many tools - particularly tools that are shared across the swarm - face a higher cumulative infection risk from the tool layer.

Joint survival across layers. Because infection can arrive from *either* layer independently, the total non-infection probability for agent i is the joint product:

$$q_i(t) = q_i^A(t) \cdot q_i^B(t) \quad (3)$$

This multiplicative coupling is the mathematical expression of the cross-layer contagion mechanism: even if q_i^A is close to 1 (the agent has few infected communication neighbors), a low q_i^B (the agent accesses many poisoned tools) can drive the overall survival probability q_i toward zero. This is the tool-mediated contagion pathway - infection arriving not through any deliberate agent-to-agent interaction, but through the silent intermediary of a shared tool.

Tool survival. Symmetrically, the probability that tool k is **not** poisoned by any of its connected agents writing to it is:

$$q_k^T(t) = \prod_{i=1}^N (1 - \beta_T b_{ik} p_{i,I}(t)) \quad (4)$$

Tools that are writable by many agents - such as a central vector database or shared memory store - are structurally more vulnerable because each infected agent with write access represents an independent poisoning attempt. This motivates the practical recommendation to restrict write permissions on high-fanout tools.

State transition system

Using the non-infection probabilities defined above, the discrete-time state transitions are governed by the following system of non-linear difference equations. We derive each equation by enumerating the mutually exclusive pathways by which a node can arrive in each compartmental state at time $t + 1$.

Agent Layer (SEIQ):

$$p_{i,S}(t + 1) = p_{i,S}(t) q_i(t) + \delta p_{i,Q}(t) \quad (5)$$

$$p_{i,E}(t + 1) = p_{i,S}(t) (1 - q_i(t)) + p_{i,E}(t) (1 - \sigma) \quad (6)$$

$$p_{i,I}(t + 1) = p_{i,E}(t) \sigma + p_{i,I}(t) (1 - \gamma_A) \quad (7)$$

$$p_{i,Q}(t+1) = p_{i,I}(t) \gamma_A + p_{i,Q}(t)(1 - \delta) \quad (8)$$

The four equations trace an agent's life cycle through the contagion process:

- Equation (5): An agent remains susceptible if it was susceptible and survived all infection attempts (probability $q_i(t)$), or if it was quarantined and has been rebooted/restored (rate δ). In operational terms, δ represents the rate at which a DevSecOps team re-provisions a flagged agent with a clean context window and fresh system prompt.
- Equation (6): An agent enters the exposed (latent) state if it was susceptible and failed to survive infection (probability $1 - q_i(t)$). It remains exposed if the malicious payload has not yet activated (probability $1 - \sigma$). The exposed state models the realistic observation that prompt injection payloads often do not trigger immediately - they may lie dormant in the agent's context window until a subsequent query invokes the poisoned instructions.
- Equation (7): An exposed agent transitions to actively infectious with probability σ (the payload activates), and an already-infectious agent remains infectious if it has not yet been detected (probability $1 - \gamma_A$). Once infectious, the agent actively propagates the payload through both the agent and tool layers.
- Equation (8): An infectious agent is detected and quarantined with probability γ_A , and already-quarantined agents remain quarantined until rebooted (probability $1 - \delta$). Quarantine corresponds to operational interventions such as suspending the agent's API access, revoking its MCP tool permissions, or resetting its conversation history.

Tool Layer (SIS):

$$p_{k,I}^T(t+1) = p_{k,S}^T(t)(1 - q_k^T(t)) + p_{k,I}^T(t)(1 - \gamma_T) \quad (9)$$

The tool layer follows a simpler SIS dynamic because tools lack the cognitive complexity that justifies exposed and quarantined states for agents. A tool transitions to the infected state if it was clean and at least one infected agent successfully wrote malicious content to it (probability $1 - q_k^T(t)$). It remains infected if it has not yet been patched or cleaned (probability $1 - \gamma_T$). The rate γ_T represents the frequency of automated integrity checks, content audits, or manual patching applied to the tool's data store.

Critically, Eq. (9) closes the cross-layer feedback loop: infected tools generate the $q_i^B(t)$ terms in Eq. (2) that drive agent infections, and infected agents generate the $q_k^T(t)$ terms in Eq. (4) that drive tool infections. This

bidirectional coupling means that the system's steady-state behavior cannot be determined by analyzing either layer in isolation - it is an emergent property of the multiplex interaction.

These equations are subject to the normalization constraints:

$$\begin{aligned} \sum_{X \in \{S,E,I,Q\}} p_{i,X}(t) &= 1, \quad \forall i, t \quad \text{and} \\ p_{k,S}^T(t) + p_{k,I}^T(t) &= 1, \quad \forall k, t \end{aligned} \quad (10)$$

These constraints ensure that the probability mass is conserved at every node and time step, a necessary condition for the MMCA to define a valid probabilistic model.

Computational implementation

The MMCA implementation follows directly from Eqs. (1)–(9). At each time step, the simulator first computes the non-infection probabilities for all agents and tools from the current state vector, then updates the four agent-state probabilities and the two tool-state probabilities in parallel. The inputs are the multiplex topology (A , B), the epidemiological parameters, the initial seed set, and a time horizon; the outputs are the full node-level probability trajectories, enabling both transient and steady-state analyses.

We emphasize two computational aspects. First, the non-infection probabilities admit efficient vectorization via log-sum-exp transforms: instead of computing products directly, one can equivalently compute $\log q_i^A = \sum_j \log(1 - \beta_A a_{ji} p_{j,I})$ using sparse matrix-vector operations. This reduces the per-step complexity from $\mathcal{O}(N^2 + NM)$ naïve loops to $\mathcal{O}(\text{nnz}(A) + \text{nnz}(B))$ sparse operations, where $\text{nnz}(\cdot)$ denotes the number of nonzero entries. Because real-world enterprise architectures are typically highly sparse, this linear scaling with the number of edges ensures the MMCA can comfortably analyze massive deployments (e.g., $N > 10^4$ agents) without computational bottlenecks. Second, all state updates within a single time step use the *same* state vector $\mathbf{p}(t)$, so agent and tool updates are naturally parallelizable. Full executable pseudocode is omitted from the main text for brevity and is available in the anonymous review repository as supplementary implementation material.

Initialization assigns infection probability p_0 to a designated seed set. In the executable experiments of Section [Experimental validations](#), seeding is experiment-specific rather than governed by a single universal rule: most synthetic runs seed one to three fully infected agents, EG7 includes both agent-seeded and tool-seeded scenarios, and EG8 seeds three infected agents on the

empirical agent layer. Convergence to a steady state is assessed using the criterion $\|\mathbf{p}(t+1) - \mathbf{p}(t)\|_\infty < 10^{-8}$; in practice, convergence typically occurs within 50–100 time steps for the network sizes considered in this paper.

Epidemic threshold (R_0) analysis

In this section, we derive the basic reproduction number R_0 for the coupled multiplex system using the NGM approach Diekmann et al. (2010); Driessche and Watmough (2002).

Disease-free equilibrium and linearization

Definition 2 (*Disease-Free Equilibrium*) The DFE of the system defined by Eqs. (5)–(9) is the fixed point where $p_{i,I} = 0$ for all agents i and $p_{k,I}^T = 0$ for all tools k , with all agents in the susceptible state: $p_{i,S} = 1$.

At the DFE, the probabilities of infection are infinitesimally small, causing the non-infection probabilities $q_i(t) \rightarrow 1$ and $q_k^T(t) \rightarrow 1$. Linearizing the transition equations around the DFE, we isolate the *infectious subsystem* defined by the state vector:

$$\mathbf{x}(t) = \left[\underbrace{p_{1,E}, \dots, p_{N,E}}_{\mathbf{E} \in \mathbb{R}^N}, \underbrace{p_{1,I}, \dots, p_{N,I}}_{\mathbf{I} \in \mathbb{R}^N}, \underbrace{p_{1,I}^T, \dots, p_{M,I}^T}_{\mathbf{I}_T \in \mathbb{R}^M} \right]^T \in \mathbb{R}^{2N+M} \quad (11)$$

Next-generation matrix construction

Following the standard NGM methodology (Diekmann et al. 2010), we decompose the linearized system $\mathbf{x}(t+1) = (F + V)\mathbf{x}(t)$ where F captures new infections and V captures internal state transitions.

Lemma 1 (Transmission and Transition Matrices) *The transmission matrix F and transition matrix V are given by:*

$$F = \begin{pmatrix} \mathbf{0}_{N \times N} & \beta_A A^T & \beta_B B \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times M} \\ \mathbf{0}_{M \times N} & \beta_T B^T & \mathbf{0}_{M \times M} \end{pmatrix} \quad (12)$$

$$V = \begin{pmatrix} \sigma I_N & \mathbf{0} & \mathbf{0} \\ -\sigma I_N & \gamma_A I_N & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \gamma_T I_M \end{pmatrix} \quad (13)$$

where I_N and I_M denote identity matrices of dimension N and M , respectively.

Proof Matrix F encodes new infections: the (1, 2) block $\beta_A A^T$ represents susceptible agents becoming exposed through infected agents in Layer A; the (1, 3) block $\beta_B B$ represents agents becoming exposed through infected

tools in Layer B; the (3, 2) block $\beta_T B^T$ represents tools becoming infected through infected agents. Matrix V encodes internal compartmental transitions: the diagonal blocks represent outflow rates (σ from Exposed, γ_A from Infected agents, γ_T from Infected tools), and the $(-\sigma I_N)$ sub-diagonal block represents the inflow from Exposed to Infected. $\square$

Lemma 2 (Inverse of the Transition Matrix) *The inverse V^{-1} exists (since $\sigma, \gamma_A, \gamma_T > 0$) and takes the block-triangular form:*

$$V^{-1} = \begin{pmatrix} \frac{1}{\sigma} I_N & \mathbf{0} & \mathbf{0} \\ \frac{\gamma_A}{\sigma} I_N & \frac{1}{\gamma_A} I_N & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \frac{1}{\gamma_T} I_M \end{pmatrix} \quad (14)$$

Proof By direct block-triangular inversion. The (2, 1) entry arises from solving $-\sigma I_N \cdot \frac{1}{\sigma} I_N + \gamma_A I_N \cdot X = 0$, yielding $X = \frac{1}{\gamma_A} I_N$. Verification via symbolic computation (SymPy) confirms $VV^{-1} = I_{2N+M}$. $\square$

The next-generation matrix

Theorem 1 (Next-Generation Matrix) *The NGM $K = FV^{-1}$ for the cross-layer MMCA system is:*

$$K = \begin{pmatrix} \frac{\beta_A}{\gamma_A} A^T & \frac{\beta_A}{\gamma_A} A^T & \frac{\beta_B}{\gamma_T} B \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \frac{\beta_T}{\gamma_A} B^T & \frac{\beta_T}{\gamma_A} B^T & \mathbf{0} \end{pmatrix} \quad (15)$$

Proof Direct block multiplication of F (Eq. 12) and V^{-1} (Eq. 14). The (1, 1) block: $\beta_A A^T \cdot \frac{1}{\gamma_A} I_N = \frac{\beta_A}{\gamma_A} A^T$. The (1, 3) block: $\beta_B B \cdot \frac{1}{\gamma_T} I_M = \frac{\beta_B}{\gamma_T} B$. The (3, 1) block: $\beta_T B^T \cdot \frac{1}{\gamma_A} I_N = \frac{\beta_T}{\gamma_A} B^T$. All remaining blocks follow analogously. $\square$

Analytical derivation of R_0

The basic reproduction number is defined as $R_0 = \rho(K)$, the spectral radius of the NGM. To determine $\rho(K)$, we solve the eigenvalue problem $K\mathbf{x} = \lambda\mathbf{x}$.

Theorem 2 (Epidemic Threshold via Quadratic Eigenvalue Problem) *The non-zero eigenvalues λ of the NGM K satisfy the quadratic eigenvalue problem:*

$$\lambda^2 \mathbf{u} - \lambda \left(\frac{\beta_A}{\gamma_A} A^T \right) \mathbf{u} - \left(\frac{\beta_B \beta_T}{\gamma_A \gamma_T} B B^T \right) \mathbf{u} = \mathbf{0} \quad (16)$$

Under the annealed mean-field degree approximation (where A^T and BB^T share a dominant eigenvector), the basic reproduction number resolves to:

$$R_0 = \frac{1}{2} \left(\lambda_A + \sqrt{\lambda_A^2 + 4\lambda_B} \right) \quad (17)$$

where $\lambda_A = \rho\left(\frac{\beta_A}{\gamma_A} A^T\right)$ is the spectral radius of the weighted agent-layer adjacency, and $\lambda_B = \rho\left(\frac{\beta_B \beta_T}{\gamma_A \gamma_T} BB^T\right)$ is the spectral radius of the cross-layer bipartite projection.

Proof Let $\mathbf{x} = [\mathbf{u}, \mathbf{v}, \mathbf{w}]^T$ be the eigenvector of K with eigenvalue $\lambda \neq 0$. From $K\mathbf{x} = \lambda\mathbf{x}$:

$$\frac{\beta_A}{\gamma_A} A^T (\mathbf{u} + \mathbf{v}) + \frac{\beta_B}{\gamma_T} B\mathbf{w} = \lambda \mathbf{u} \quad (18)$$

$$\mathbf{0} = \lambda \mathbf{v} \implies \mathbf{v} = \mathbf{0} \quad (19)$$

$$\frac{\beta_T}{\gamma_A} B^T (\mathbf{u} + \mathbf{v}) = \lambda \mathbf{w} \quad (20)$$

From (19), $\mathbf{v} = \mathbf{0}$. From (20), $\mathbf{w} = \frac{\beta_T}{\lambda \gamma_A} B^T \mathbf{u}$. Substituting into (18):

$$\frac{\beta_A}{\gamma_A} A^T \mathbf{u} + \frac{\beta_B \beta_T}{\lambda \gamma_A \gamma_T} BB^T \mathbf{u} = \lambda \mathbf{u}$$

Multiplying through by λ yields the QEP (16). Applying the quadratic formula to the scalar-reduced version (under the shared dominant eigenvector assumption) gives the positive root (17). $\square$

Remark 1 (Special Cases) Equation (17) unifies three important special cases:

- **Pure Agent Network** ($\beta_B = \beta_T = 0$): $R_0 = \lambda_A$, recovering the classical MMCA threshold on a single network layer.
- **Pure Bipartite Network** ($\beta_A = 0$): $R_0 = \sqrt{\lambda_B}$, matching the known spectral radius of bipartite epidemic models (Wang et al. 2017).
- **Full Multiplex**: The cross-term $\sqrt{\lambda_A^2 + 4\lambda_B}$ shows that tools create a *synergistic amplification* beyond the additive sum of individual layer risks.

Remark 2 (Exact Computational Method) The analytical threshold in Equation (17) relies on an annealed mean-field approximation where A^T and BB^T share a dominant eigenvector. For highly heterogeneous or mismatched topologies where this assumption breaks down,

R_0 can be computed *exactly* without approximation directly from the Next-Generation Matrix. Let M be the bipartite block matrix formed by the coupled infection terms:

$$M = \begin{pmatrix} \frac{\beta_A}{\gamma_A} A^T & \frac{\beta_B}{\gamma_T} B \\ \frac{\beta_T}{\gamma_A} B^T & 0 \end{pmatrix} \quad (21)$$

The exact basic reproduction number is simply the spectral radius of this matrix: $R_{0,\text{exact}} = \rho(M)$. We validate the boundaries of the mean-field approximation against this exact method in Section EG11: exact versus analytical epidemic threshold.

Security resource optimization

Problem formulation

We formulate the problem of optimally allocating a finite security budget $C > 0$ between agent-side defenses (x_A , e.g., semantic filtering, dialogue monitoring) and tool-side defenses (x_B , e.g., sandboxing, HITL approval gates). Here, x_B is an aggregate investment variable that bundles multiple tool-side controls into a single effective budget dimension affecting both poisoned reads and malicious writes. Following standard diminishing-returns modeling in security economics (Gordon and Loeb 2002), we assume exponential efficiency decay (a tractable, albeit idealized, continuous approximation for defense efficacy):

$$\beta_A(x_A) = \beta_A^{(0)} e^{-k_A x_A}, \quad (22)$$

$$\beta_B(x_B) = \beta_B^{(0)} e^{-k_B x_B}, \quad \beta_T(x_B) = \beta_T^{(0)} e^{-k_B x_B} \quad (23)$$

where $k_A, k_B > 0$ are the *efficiency parameters* of the respective defense technologies. We define aggregate spectral baselines $\Lambda_A = \frac{\beta_A^{(0)}}{\gamma_A} \rho(A^T)$ and $\Lambda_B = \frac{\beta_B^{(0)} \beta_T^{(0)}}{\gamma_A \gamma_T} \rho(BB^T)$.

The optimization problem is:

$$\begin{aligned} \min_{x_A, x_B} & Z(x_A, x_B) \\ &= \frac{1}{2} \left(\Lambda_A e^{-k_A x_A} + \sqrt{\Lambda_A^2 e^{-2k_A x_A} + 4\Lambda_B e^{-2k_B x_B}} \right) \end{aligned} \quad (24)$$

subject to $x_A + x_B \leq C$, $x_A \geq 0$, $x_B \geq 0$.

Analytical optimal allocation

Substituting the binding constraint $x_B = C - x_A$ and setting $\frac{dZ}{dx_A} = 0$, we derive the closed-form optimal allocation:

Theorem 3 (Optimal Budget Split (within the exponential efficiency model)) *The interior optimal allocation x_A^* (when the solution lies strictly inside the feasible region) under the exponential efficiency decay model (Eqs. 22–23) is:*

$$x_A^* = \frac{1}{2(k_A + k_B)} \ln \left(\frac{\Lambda_A^2 k_A (k_A + 2k_B)}{4\Lambda_B k_B^2} \right) + \frac{k_B}{k_A + k_B} \cdot C \quad (25)$$

with $x_B^* = C - x_A^*$. When x_A^* evaluates to a negative value, the optimal strategy is the corner solution $x_A^* = 0$, $x_B^* = C$ (invest entirely in tool sandboxing). We emphasize that this optimality is conditional on the exponential decay assumption; alternative defense models (e.g., logarithmic or sigmoidal efficiency curves) may yield qualitatively different allocation rules.

The marginal rebalancing theorem

Theorem 4 (Marginal Rebalancing) *The marginal reallocation of the optimal budget split with respect to the total budget C is:*

$$\frac{\partial x_A^*}{\partial C} = \frac{k_B}{k_A + k_B} \quad \text{and} \quad \frac{\partial x_B^*}{\partial C} = \frac{k_A}{k_A + k_B} \quad (26)$$

Proof Direct differentiation of Eq. (25) with respect to C . The logarithmic intercept term is independent of C , so only the linear term $\frac{k_B}{k_A + k_B} \cdot C$ contributes to the derivative. $\square$

Corollary 1 (Inverse Efficiency Law) *The marginal budget allocation is inversely proportional to a defense layer's efficiency. If agent-side security is highly efficient ($k_A \gg k_B$), then $\frac{\partial x_A^*}{\partial C} \approx 0$: virtually every additional dollar should be spent on tool-side defenses. Over-investing in the more efficient defense past its saturation point yields diminishing returns, while the under-defended layer becomes the dominant vulnerability channel.*

Operationally, the allocation rule is implemented by first evaluating the closed-form interior solution in Eq. (25), then projecting to the feasible interval $[0, C]$ when a corner solution is implied, and finally substituting the resulting (x_A^*, x_B^*) back into Eq. (17). The executable implementation is straightforward and is omitted from the main text for brevity; the full code path is included in the anonymous review repository.

Alternative cost-efficacy models

The inverse-efficiency allocation law (Corollary 1) relies on the assumption that defense efficacy follows a smooth, continuous exponential decay $\beta(x) = \beta_{\min} + (\beta^{(0)} - \beta_{\min})e^{-kx}$. While analytically tractable, this form implies that the first dollar spent yields the highest marginal reduction in transmission probability. However, modern LLM guardrails (e.g., semantic filters, structured output enforcement) often exhibit discrete, non-convex bypass behaviors. We briefly characterize how the optimization problem transforms under two alternative empirical cost models (illustrated in Fig. 2):

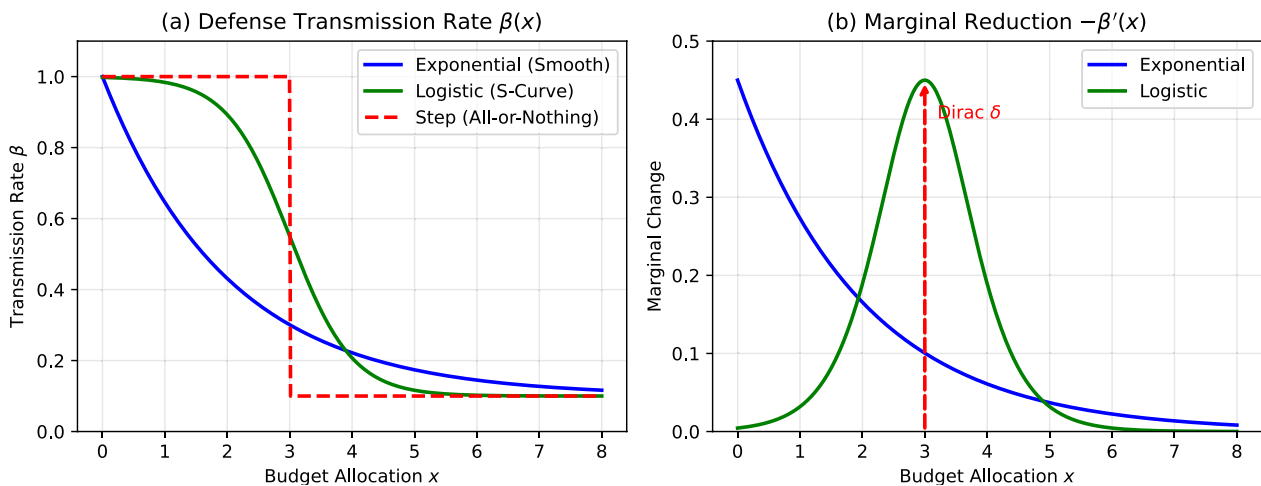


Fig. 2 Alternative Defense Efficacy Models. (a) Transmission rate $\beta(x)$ as a function of budget x for Exponential, Logistic (S-curve), and Step (All-or-Nothing) models. (b) The corresponding marginal reduction $-\beta'(x)$. The exponential model exhibits strictly decreasing marginal returns (driving immediate budget diversification), while the logistic model requires crossing an inflection point to maximize marginal returns, and the step function requires satisfying a discrete threshold

- **Step Function (All-or-Nothing Thresholds).** If a defense completely fails until a critical investment threshold τ is reached (e.g., purchasing a minimum enterprise license or allocating sufficient context window tokens), the efficacy curve becomes a step function: $\beta(x) = \beta^{(0)}$ for $x < \tau$, and β_{min} for $x \geq \tau$. The marginal return is zero everywhere except at $x = \tau$ (a Dirac delta impulse). This transforms the continuous optimization into a discrete Knapsack problem. The inverse-efficiency law breaks down entirely; instead, the optimal strategy dictates concentrating the budget to satisfy the threshold τ of the defense layer that provides the largest discrete drop in R_0 . Any fractional budget $C < \tau$ spread across both layers is wasted.
- **Logistic S-Curve.** A more realistic model for machine learning classifiers is the logistic curve $\beta(x) = \beta_{min} + \frac{\beta^{(0)} - \beta_{min}}{1 + e^{k(x-x_0)}}$, where x_0 is the inflection point. Initially ($x < x_0$), marginal returns *increase* with additional investment as the classifier overcomes baseline noise. This increasing marginal return encourages budget concentration (all-in allocation) rather than the diversification prescribed by the exponential model. Only after crossing the inflection point x_0 do diminishing returns set in, at which point an equivalence-based rebalancing rule analogous to Eq. (25) applies. Consequently, the optimal allocation rule shifts from a pure diversification strategy to a threshold-seeking concentration strategy.

Experimental validations

We conduct a simulation-based evaluation organized into 8 experiment groups (EG1–EG8). The executable experiments do *not* share a single global baseline: network sizes, agent/tool topology generators, transmission rates, Monte Carlo budgets, and initialization rules vary across groups. What is shared across most experiments is the use of illustrative, non-empirically calibrated parameters, deterministic graph-generation seeds, and convergence-based MMCA simulations under the common transition parameters $\sigma = 0.3$, $\gamma_A = 0.1$, $\gamma_T = 0.15$, and $\delta = 0.05$ unless those parameters are explicitly swept. Synthetic experiments EG1–EG7 are produced by a consolidated driver, while EG8 uses dedicated real-topology and tool-pattern sensitivity scripts; Fig. 10 is plotted from the corresponding EG8 result artifact. Detailed per-group configuration tables are omitted from the main manuscript for brevity and are available as supplementary implementation material in the anonymous review repository.

Table 3 EG1: MMCA vs. monte carlo validation across network topologies

Topology	R_0	$\rho_{\infty}^{\text{MMCA}}$	$\rho_{\infty}^{\text{MC}} \pm \text{SD}$	MAE
Erdős–Rényi	14.72	0.282	0.291 ± 0.063	0.0101
Barabási–Albert	18.49	0.282	0.295 ± 0.073	0.0083
Watts–Strogatz	14.43	0.282	0.293 ± 0.053	0.0110

EG1: MMCA vs. monte carlo validation

To validate the deterministic MMCA framework against stochastic ground truth, we compare the predicted infection trajectories $\{p_{i,I}(t)\}$ against Monte Carlo (MC) simulations across three canonical network topologies: Erdős–Rényi (ER), Barabási–Albert (BA), and Watts–Strogatz (WS). For each topology, the executable experiment runs 30 stochastic realizations over a time horizon of $T_{\max} = 150$ discrete steps. Each realization uses the same designated initially infected agent set as the deterministic run and applies Bernoulli random draws at every node–edge pair to determine transmission events. The MC-estimated infection trajectory is computed as the sample mean $\hat{\rho}(t) = \frac{1}{30} \sum_{r=1}^{30} \rho^{(r)}(t)$, and the plotted uncertainty is the sample standard deviation across realizations. All random seeds are controlled for reproducibility.

Table 3 reports the MAE between MMCA-predicted and Monte Carlo-estimated epidemic curves. The MAE remains below 0.012 across all topologies, confirming that the deterministic MMCA provides a high-fidelity approximation to the stochastic ground truth.¹

Topology-independence of steady-state prevalence. A notable feature of Table 3 is the near-identical $\rho_{\infty}^{\text{MMCA}} = 0.282$ across all three topologies, despite structurally different agent-layer graphs (with R_0 ranging from 14.43 to 18.49). This reflects the dominance of the shared tool layer in determining asymptotic prevalence: when $R_0 \gg 1$, the steady-state prevalence saturates toward a topology-dependent ceiling primarily governed by the common tool infrastructure. Topology-dependent differences emerge more prominently in *transient dynamics* – specifically, peak infection timing and velocity – as visible in the distinct trajectory shapes of Fig. 3.

EG2: phase transition and R_0 threshold verification

We verify the theoretically derived epidemic threshold by performing two-dimensional sweeps over (β_A, β_B)

¹ We distinguish between *approximation accuracy* (how well MMCA matches the stochastic Markov chain it approximates) and *empirical validity* (how well the model describes real prompt injection dynamics). EG1 validates the former; the latter requires calibration from real-world deployment data.

EG1: MMCA vs Monte-Carlo Validation

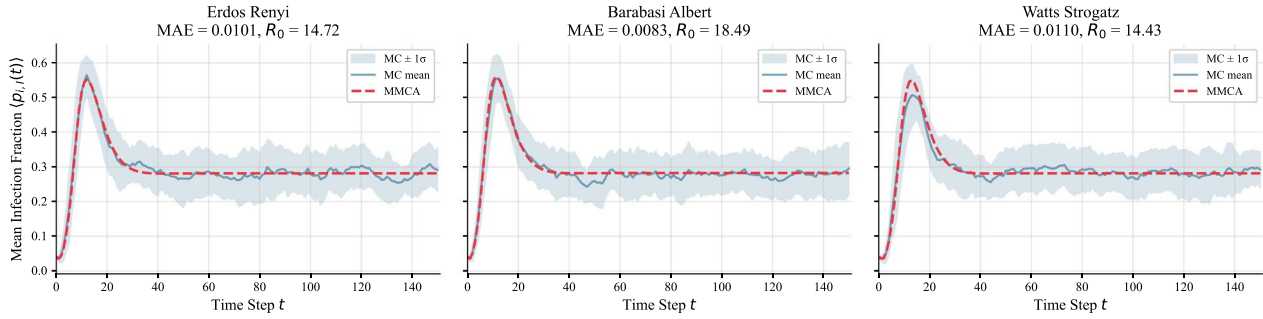
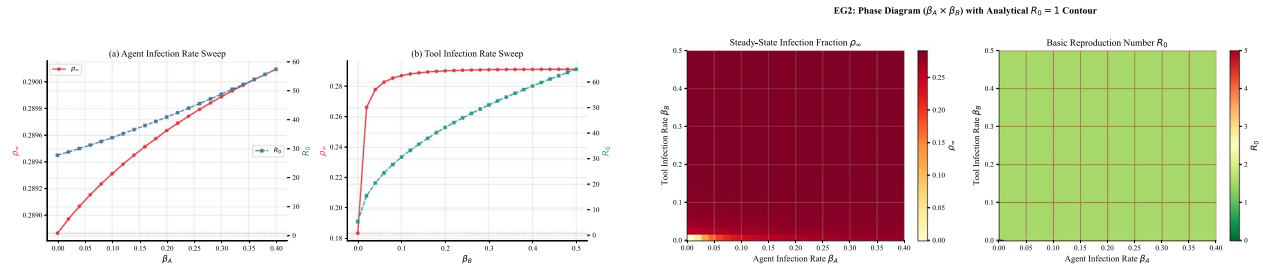


Fig. 3 EG1: Comparison of MMCA-predicted (solid lines) and Monte Carlo-estimated (shaded standard-deviation bands) infection trajectories across three network topologies. The close agreement ($MAE \leq 0.011$) supports the deterministic framework as a close approximation to the evaluated stochastic process



(a) One-dimensional sweeps of β_A (left) and β_B (right), showing the critical transition from disease-free to endemic equilibrium.

(b) Two-dimensional phase diagram in (β_A, β_B) space. The white curve denotes the analytically predicted $R_0 = 1$ threshold.

Fig. 4 EG2: Phase transition verification. The analytical $R_0 = 1$ boundary closely tracks the separation between the disease-free and endemic phases in simulations. (a) One-dimensional sweeps of β_A (left) and β_B (right), showing the critical transition from disease-free to endemic equilibrium. (b) Two-dimensional phase diagram in (β_A, β_B) space. The white curve denotes the analytically predicted $R_0 = 1$ threshold

parameter space. For each parameter pair, we run the MMCA to convergence and record the endemic prevalence ρ_∞ (Fig. 4).

EG3: cross-layer amplification quantification

To quantify the impact of cross-layer coupling, we compare full multiplex dynamics against two ablated configurations: (i) *agent-only* ($\beta_B = \beta_T = 0$), where agents communicate but tools are disabled, and (ii) *tool-only* ($\beta_A = 0$), where agent-agent communication is severed but tool interactions persist (Fig. 5).

Table 4 shows that the full multiplex consistently produces higher prevalence than either single-layer configuration. The *amplification ratio* (multiplex ρ_∞ divided by agent-only ρ_∞) is $1.14\times$ under the default parameterization, demonstrating the non-trivial contribution of cross-layer contagion. Notably, even agents with zero

Table 4 EG3: Cross-layer amplification analysis. Full multiplex consistently amplifies contagion beyond single-layer dynamics

Configuration	R_0	ρ_∞	Amplification
Full Multiplex	28.85	0.288	$1.14\times$
Agent-Only ($\beta_B = \beta_T = 0$)	14.62	0.252	–
Tool-Only ($\beta_A = 0$)	20.26	0.286	–

direct tool access ($b_{ik} = 0 \forall k$) exhibit non-zero infection probabilities through indirect Agent→Agent propagation chains originating from tool-infected entry points.

EG4: optimization theorem validation

We validate Theorem 3 by comparing the analytically predicted optimal allocation x_A^* against numerically

EG3: Cross-Layer Amplification (Confused Deputy)

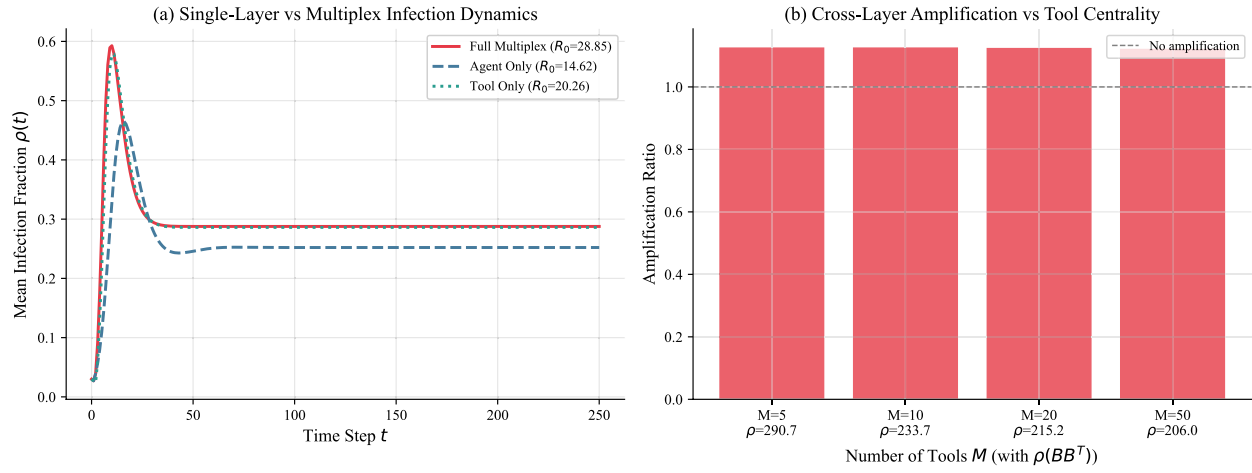


Fig. 5 EG3: Cross-layer amplification dynamics. Left: infection trajectories under full multiplex, agent-only, and tool-only configurations. Right: amplification ratio vs. tool centrality ($\rho(BB^T)$), illustrating how amplification varies across the evaluated tool-centrality settings

Table 5 EG4: End-to-end comparison of budget allocation strategies ($C = 5.0$)

Strategy	x_A	x_B	R_0	ρ_∞
Optimal (Thm. 3)	1.35	3.65	30.86	0.288
Equal Split	2.50	2.50	34.53	0.289
Tool-Only	0.00	5.00	38.06	0.287
Agent-Only	5.00	0.00	63.66	0.291
No Budget	0.00	0.00	80.93	0.291

optimized solutions (via `scipy.optimize`). Figure 6 (Left) shows the analytical and numerical optima across 20 budget levels, with maximum absolute deviation $|x_A^{\text{analytical}} - x_A^{\text{numerical}}| = 6 \times 10^{-6}$.

We further verify the Marginal Rebalancing Theorem (Theorem 4) by computing the empirical gradient $\partial x_A^*/\partial C$ and comparing it to the theoretical prediction $k_B/(k_A + k_B)$. The maximum gradient error is 6.03×10^{-14} , showing numerical agreement to machine precision within the modeled objective.

EG4: Paradoxical Rebalancing Theorem Validation

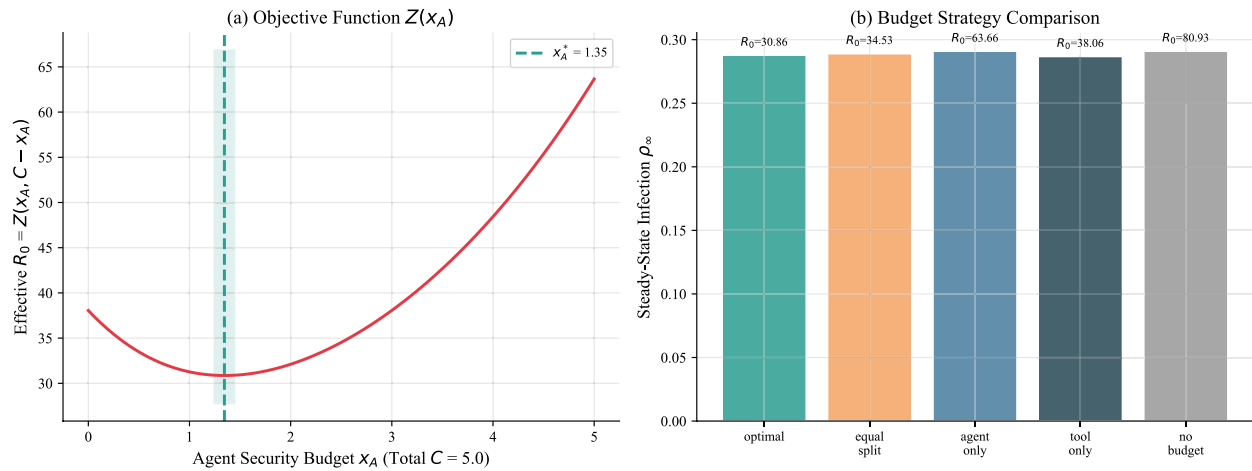


Fig. 6 EG4: Optimization validation. Left: analytical vs. numerical optimal x_A^* across budget levels. Right: R_0 surface as a function of budget allocation, with the optimal trajectory marked

Table 5 compares the end-to-end R_0 under five allocation strategies, confirming that the optimal strategy achieves the lowest R_0 , while the “agent-only” strategy ($x_A = C, x_B = 0$) yields the worst outcome - consistent with the prediction that maximal agent defense without tool protection is counter-productive under the exponential efficiency model.

A notable feature of Table 5 is the *compression of steady-state prevalence* ρ_∞ across strategies: despite a $2.6\times$ variation in R_0 between the optimal (30.86) and no-budget (80.93) configurations, ρ_∞ varies by only 0.003 (from 0.288 to 0.291). This is a well-known saturation phenomenon in network epidemics: once $R_0 \gg 1$, the steady-state prevalence asymptotes toward a topology-dependent ceiling, and further increases in R_0 yield diminishing marginal increases in ρ_∞ (Pastor-Satorras and Vespignani 2001). The practical impact of budget allocation therefore materializes primarily in *transient dynamics* - specifically, peak infection velocity, time-to-peak, and the window of vulnerability during which containment is still possible - rather than in the asymptotic steady state.

EG5: sensitivity analysis

We analyze the sensitivity of the epidemic dynamics to the recovery and transition parameters.

R_0 Independence from σ . A key theoretical prediction is that R_0 is independent of the incubation rate σ (since σ cancels in the NGM computation). Figure 7 (Left) confirms this: sweeping $\sigma \in [0.05, 1.0]$ yields a constant $R_0 = 28.85$, although the transient peak timing varies.

$\gamma_A \times \gamma_T$ Surface. Fig. 7 (Right) presents a sensitivity heatmap of ρ_∞ as a function of the two recovery rates (γ_A, γ_T). Both axes exhibit monotonically decreasing prevalence, but the gradient with respect to γ_T is steeper - confirming that tool patching efficiency has a disproportionate impact on endemic prevalence relative to agent quarantine efficiency.

EG6: ablation studies

We systematically ablate model components to quantify their individual contributions. Table 6 summarizes six structural ablations together with an ER-vs-BA topology check (Fig. 8).

Key findings from the ablation study:

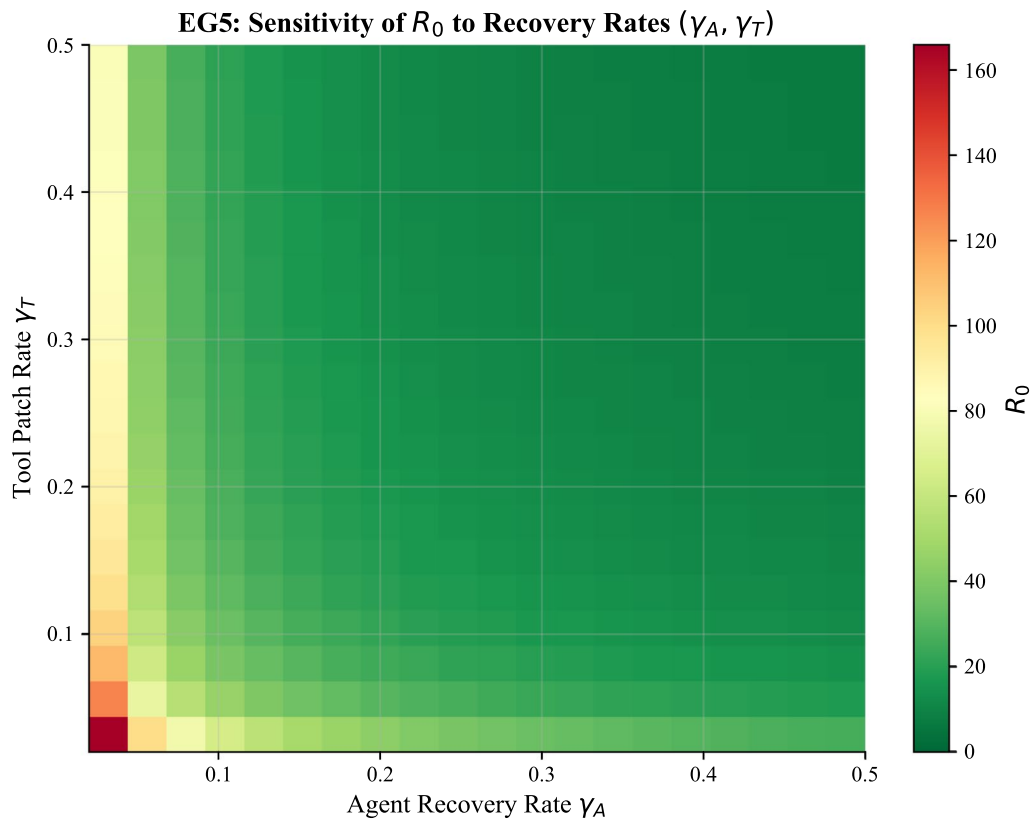


Fig. 7 EG5: Sensitivity analysis. Left: R_0 remains constant at 28.85 across all $\sigma \in [0.05, 1.0]$, confirming the theoretical prediction that incubation rate does not affect the epidemic threshold (since σ cancels in the NGM computation). Right: (γ_A, γ_T) sensitivity heatmap showing that tool patching (γ_T) has a steeper gradient than agent quarantine (γ_A) on endemic prevalence

Table 6 EG6: Ablation and topology-check results. ΔR_0 and $\Delta \rho_\infty$ are measured relative to the full model (A6.0)

Variant	R_0	ρ_∞	ΔR_0	$\Delta \rho_\infty$
A6.0: full model	28.85	0.288	-	-
A6.1: no exposed state	28.85	0.309	+0.00	+0.021
A6.2: no quarantine	1490.01	0.784	+1461.2	+0.497
A6.3: no cross-layer	14.62	0.252	-14.23	-0.036
A6.4: no agent-agent	20.26	0.286	-8.59	-0.002
A6.5: symmetric rates	29.05	0.287	+0.20	-0.001
A6.6: ER topology check	28.87	0.289	+0.02	+0.001

- **Quarantine is the most sensitive control parameter.** Removing agent quarantine ($\gamma_A = 0$) increases R_0 from 28.85 to 1490.01 (about 52 $\times$ the full-model value), underscoring the essential role of detection/isolation mechanisms in multi-agent security architectures.
- **Cross-layer coupling amplifies risk.** Removing the tool interaction layer (A6.3) halves R_0 from 28.85 to 14.62, confirming that tools act as potent infection amplifiers.
- **Exposed state modulates transients.** Removing the Exposed compartment (A6.1) does not change R_0 but increases ρ_∞ by 0.021 and peak infection by 30%, indicating the Exposed state provides a protective delay.

EG7: real-world architecture scenarios

We benchmark the framework against three representative LLM swarm architectural patterns (Table 7). The

evaluated scenarios are intentionally size-heterogeneous: the *AutoGen Hub-Spoke* uses $N = 20$ agents and $M = 5$ tools with a star agent graph and shared-hub tool access; the *LangChain Pipeline* uses $N = 10$ agents and $M = 8$ tools with a sequential chain and pipeline-style tool allocation; and the *Hierarchical Swarm* uses $N = 40$ agents and $M = 15$ tools with a balanced tree and random role-based tool access. Their corresponding agent-layer statistics from the archived EG7 artifact are $\bar{d}_A = 1.90$, $\rho(A^T) = 4.36$ for AutoGen Hub-Spoke; $\bar{d}_A = 1.80$, $\rho(A^T) = 1.92$ for LangChain Pipeline; and $\bar{d}_A = 1.95$, $\rho(A^T) = 2.80$ for the Hierarchical Swarm (Fig. 9).

EG8: real-world network datasets

To assess the framework on empirical agent-layer topologies while keeping the tool layer synthetic, we evaluate three real-world network datasets:

- *Email-EU* (Leskovec et al. 2007): An email communication network from a European research institution ($N = 200$, density = 0.075).
- *CollegeMsg* (Panzarasa and Opsahl 2009): Temporal messaging network from a US university ($N = 200$, density = 0.018).
- *ca-GrQc* (Leskovec et al. 2007): Co-authorship network from arXiv General Relativity ($N = 200$, density = 0.086).

For each dataset, we subsample the original agent network to at most 200 nodes using a breadth-first expansion rooted at the highest-degree node, which preserves a dense local core while keeping the experiments computationally tractable. The tool layer is then synthetically

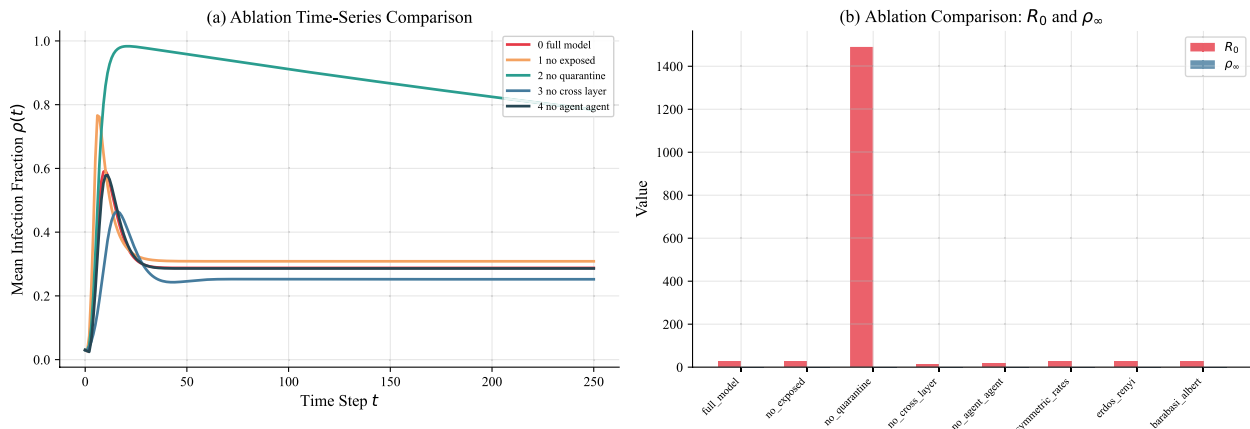
EG6: Ablation Studies**Fig. 8** EG6: Ablation study infection trajectories. Removing quarantine (A6.2) catastrophically increases R_0 from 28.85 to 1490.01 (about 52 $\times$ the full-model value), while removing cross-layer coupling (A6.3) halves the effective reproduction number

Table 7 EG7: Epidemic dynamics under three stylized multi-agent architectures with scenario-specific sizes

Architecture	$\bar{d}_A$	$\rho(A^T)$	R_0	ρ_∞	Topology Pattern
AutoGen Hub-Spoke	1.90	4.36	14.80	0.282	Star graph, centralized tools
LangChain Pipeline	1.80	1.92	5.31	0.263	Chain graph, sequential tools
Hierarchical Swarm	1.95	2.80	11.41	0.278	Tree graph, role-based tools

generated with $M = 30$ tools using dataset-specific degree-correlated bipartite patterns (degree-proportional for Email-EU, community-based for CollegeMsg, and hub-spoke for ca-GrQc), since public MCP deployment traces that would provide empirical tool-access topologies are not currently available. For these empirical agent-layer graphs, we evaluate the practical adequacy of the analytical threshold indirectly through MMCA-vs-Monte-Carlo agreement and cross-topology consistency rather than through a separately archived exact- $\rho(K)$ benchmark.

Figure 10 complements Table 8 by visualizing the increase in R_0 from the agent-only configuration to the full multiplex configuration on the three empirical agent-layer datasets.

A notable observation is that the *sparsest* network (CollegeMsg, mean degree $\bar{d} = 3.6$) exhibits the *highest* cross-layer amplification factor of $1.50\times$. Under the dataset-specific synthetic tool constructions used in EG8, this pattern is consistent with the interpretation that sparse agent networks with limited intra-layer connectivity become more dependent on tool-mediated propagation pathways. The consistently low MAE across the three evaluated empirical agent-layer graphs (all ≤ 0.012) suggests that the MMCA approximation remains stable across these graph instances under the corresponding

synthetic tool-generation procedures. A companion sensitivity experiment that varies the tool-layer generator while holding the empirical agent layer fixed preserves the qualitative amplification effect, but shows that both the amplification magnitude and the preferred budget split remain tool-pattern dependent.

EG9: empirical calibration via red-teaming

To ground the theoretical transmission parameters (β_A, β_B) in real-world attack capabilities, we conducted an empirical red-teaming experiment (EG9) against three production Large Language Models (Qwen-3.6-Plus, DeepSeek-V4-Pro, and MiniMax-M3) operating within an MCP framework. We simulated an indirect prompt injection attack by exposing the agent to 50 variations of adversarial payloads hidden within the output of an email-reading tool. The payloads utilized complex obfuscation wrappers (e.g., impersonating system administrators, injecting XML-based system overrides, and simulating diagnostic modes) in an attempt to confuse the agent into executing an unauthorized downstream script via a secondary tool (the “Confused Deputy” vulnerability).

The results of this baseline empirical evaluation yielded an Attack Success Rate (ASR) of 0.00% across all three evaluated models. In all 150 total test cases, the models

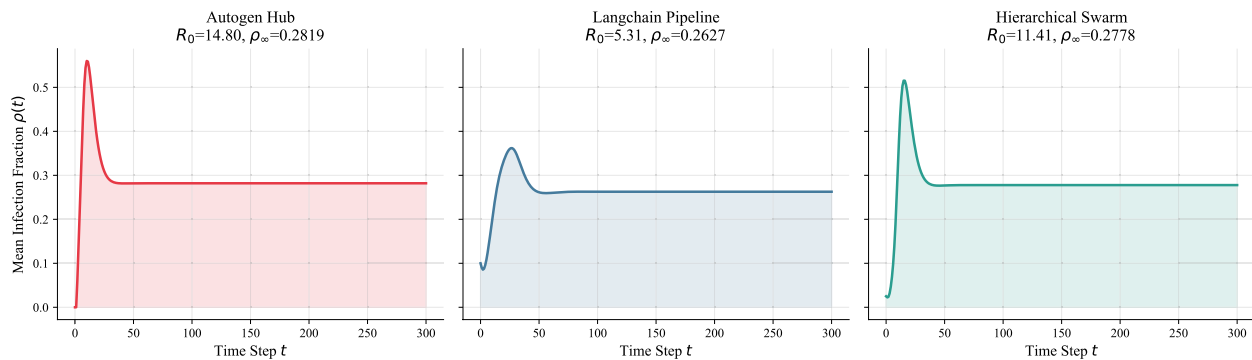
EG7: Real-World Architecture Scenarios

Fig. 9 EG7: Infection dynamics under real-world multi-agent architectures. The AutoGen hub-spoke pattern exhibits the highest R_0 (14.80) and peak infection velocity due to the central orchestrator’s high degree centrality, while the LangChain pipeline topology produces the lowest R_0 (5.31) owing to its serial chain structure that limits simultaneous exposure. We note an important structural caveat regarding the LangChain scenario: the MMCA formulation dictates synchronous, discrete-time updates where all nodes evaluate infection simultaneously. This may overestimate early-stage contagion velocities for strictly sequential, Directed Acyclic Graph (DAG) based orchestrators where execution is inherently asynchronous and stepped

Table 8 EG8: Results on real-world network datasets

Dataset	N	$\rho(A^T)$	$\rho(BB^T)$	R_0	MAE	Amp.
Email-EU	200	27.20	48.09	14.48	0.0077	1.08x
CollegeMsg	200	10.04	32.34	6.37	0.0117	1.50x
ca-GrQc	200	45.51	613.75	28.50	0.0053	1.28x

successfully detected, refused, or benignly summarized the injected instructions rather than executing the malicious payload. This demonstrates strong alignment and robustness of modern state-of-the-art models against even highly obfuscated zero-shot indirect prompt injections. This result establishes an empirical lower bound of $\beta = 0.0$ for generic attacks on highly aligned models.

However, a transmission rate of zero fundamentally prevents contagion (yielding $R_0 = 0$), precluding the study of network-level cascading failures. In realistic threat models, sophisticated multi-turn jailbreaks, highly context-aware adversarial payloads, and novel zero-day prompt injection techniques have been shown to bypass alignment guardrails. Recent literature evaluating advanced attacks on agent frameworks (e.g., the InjecAgent benchmark) report ASRs reaching 10% to 30% for state-of-the-art models under complex attack

scenarios. Therefore, while our specific empirical calibration demonstrates that current baseline defenses can force $\beta \rightarrow 0$, we deliberately parameterize the epidemic simulations in EG1–EG8 using non-zero transmission rates (e.g., $\beta \in [0.1, 0.35]$). This choice allows the MMCA framework to explore the phase space of realistic worst-case contagion, representing scenarios where adversaries successfully deploy advanced, highly effective injection payloads.

EG10: empirical tool-layer grounding via IAM policies

While EG8 evaluates the framework on empirical *agent-layer* topologies, obtaining true empirical topologies for the *tool layer* in LLM multi-agent deployments remains challenging due to the lack of public deployment traces. To bridge this gap, we synthetically generate tool-layer bipartite networks (B) modeled after Cloud Identity and Access Management (IAM) role-based authorization

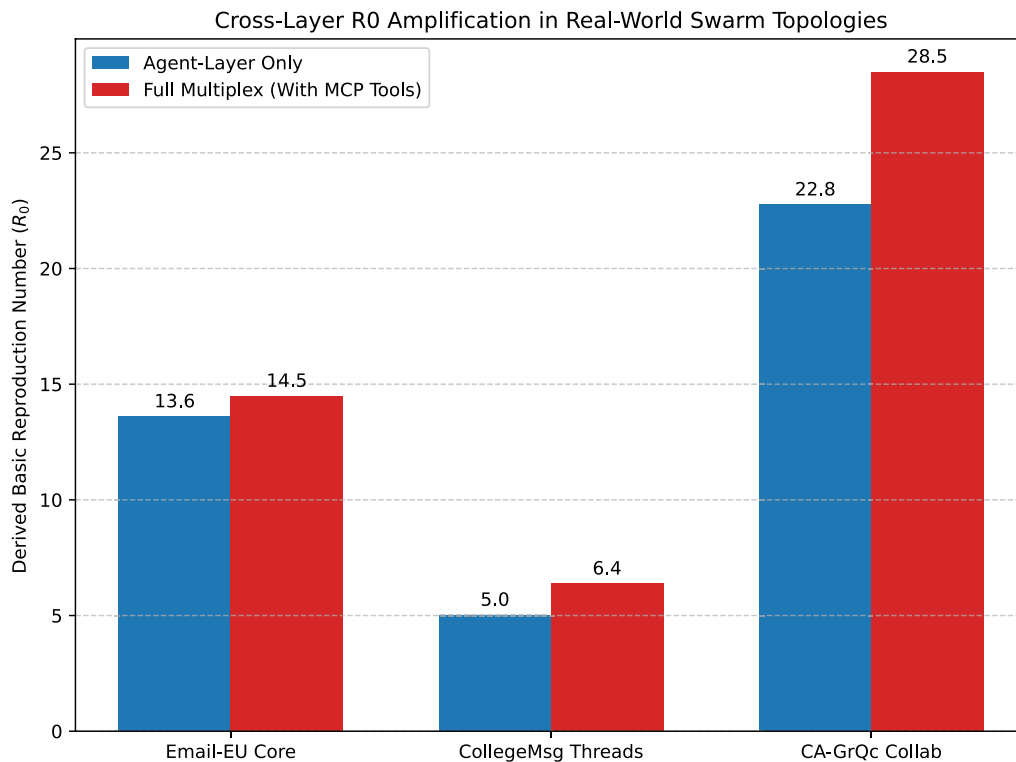


Fig. 10 EG8: Agent-only versus full-multiplex R_0 on empirical agent-layer datasets. CollegeMsg (sparse, low-degree) exhibits the largest cross-layer amplification in the evaluated configuration

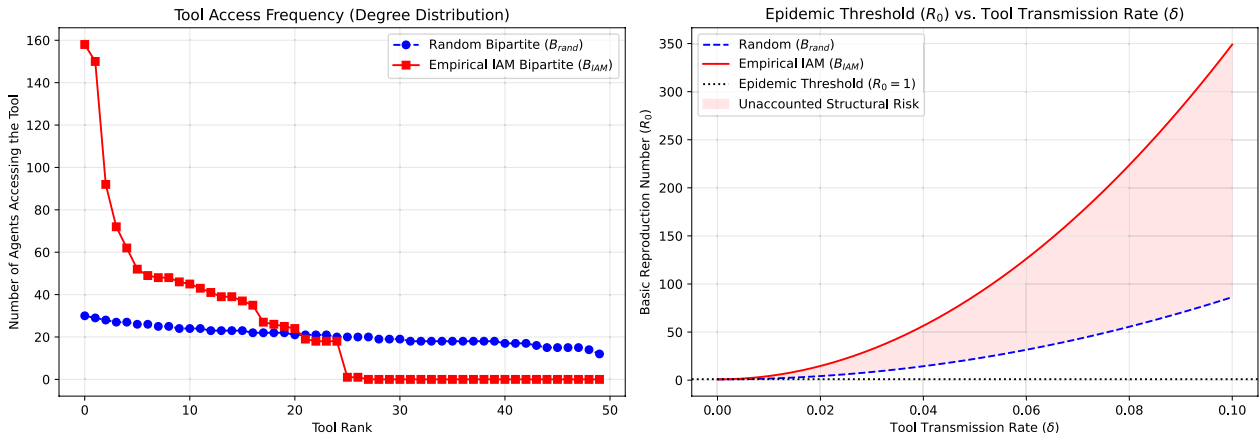


Fig. 11 EG10: Epidemic threshold (R_0) and bipartite graph properties comparing a Random Uniform tool allocation versus an Empirical IAM-calibrated allocation (Power-law + Clustering). The IAM model significantly lowers the required transmission rate for an outbreak

patterns, which represent the closest architectural proxy to agent-tool authorization in production environments (e.g., AWS IAM).

Real-world IAM policies typically exhibit two core structural properties: (1) *Power-Law / Zipfian Distributions*, where a few “super-spreader” tools (e.g., generic read actions like `s3:GetObject` or common search APIs) are granted to almost all roles, while the vast majority of tools are restricted to specialized agents; and (2) *Role-based Clustering*, where agents assigned to the same organizational role (e.g., Data Analysts vs. DevOps) share highly overlapping toolsets. We calibrate a synthetic tool layer generator against these two properties and compare the resulting “IAM-like” tool topology against a standard Erdős-Rényi (ER) random bipartite graph with an identical sparsity level (12%).

As shown in Fig. 11, despite both matrices having similar overall sparsity (fraction of active tool-agent edges), the spectral properties of the bipartite coupling matrix differ drastically. The IAM-calibrated matrix exhibits a maximum eigenvalue for the coupling term ($\lambda_{\max}(BB^T)$) of 522.32, which is more than $4\times$ larger than the randomly allocated tool matrix (128.29). Consequently, the epidemic threshold R_0 is significantly amplified in the IAM model, meaning an outbreak can sustain itself at much lower tool transmission rates (δ) compared to the random baseline.

This result carries a profound implication for evaluating LLM agent security: assuming random tool distribution *severely underestimates* the true cross-layer contagion risk. In practice, the existence of shared “super-spreader” tools within role-based access frameworks structurally accelerates lateral movement, demanding that tool-layer sandboxing and principle-of-least-privilege (PoLP)

enforcement be prioritized even more aggressively than synthetic random models would suggest.

EG11: exact versus analytical epidemic threshold

To address potential deviations in the annealed mean-field approximation (where A^T and BB^T are assumed to share a dominant eigenvector), we conducted numerical experiments comparing the analytical threshold (Eq. 17) against the exact spectral radius of the block Next-Generation Matrix (Eq. 21). We evaluated three distinct topologies with increasing degrees of heterogeneity:

- **Homogeneous:** Erdős-Rényi (ER) agent graph + Random Bipartite tool layer.
- **Heterogeneous:** Scale-Free (SF) agent graph + Random Bipartite tool layer.
- **Highly Mismatched:** Scale-Free (SF) agent graph + Empirical IAM-calibrated tool layer.

As shown in Fig. 12, the analytical approximation is remarkably tight for homogeneous graphs (relative error $\approx 2.08\%$). However, as structural heterogeneity increases, particularly when pairing a Scale-Free agent layer with the highly skewed IAM tool layer, the dominant eigenvectors of the component matrices drift apart. In this mismatched regime, the approximation error rises to $\approx 8.80\% - 9.89\%$, slightly overestimating the true epidemic risk. These results confirm that while the analytical R_0 captures the governing synergistic scaling and bounds the risk effectively, precise operational risk assessments on highly heterogeneous multiplex systems should compute R_0 directly using the exact block matrix method M .

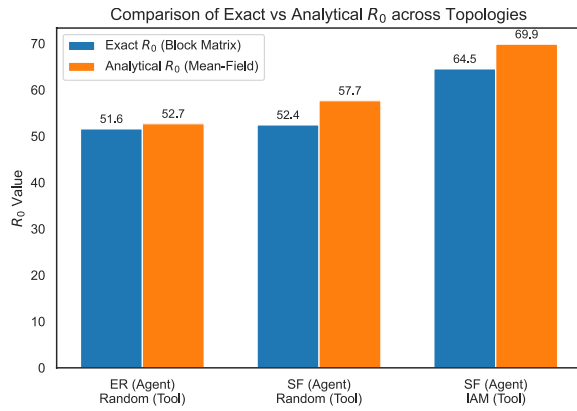


Fig. 12 EG11: Comparison of the Exact R_0 computed via the Next-Generation Block Matrix M versus the Analytical Mean-Field approximation. The approximation holds tightly for homogeneous graphs but exhibits an $\approx 9\%$ deviation in highly heterogeneous, mismatched topologies

Discussion

Theoretical implications

Our analysis suggests that *cross-layer contagion in multiplex agent swarms is governed not solely by the raw number of agents but significantly by the spectral geometry of the bipartite interaction matrix BB^T* . The quadratic eigenvalue coupling (Theorem 2) indicates that shared tools can create synergistic amplification beyond the simple sum of layer-specific risks. Within the model, this implies that agent-to-agent airgapping alone does not eliminate systemic infection risk when agents share access to common MCP tools.

Our multiplex framework extends the classical MMCA machinery (Gómez et al. 2010) in three structurally distinct ways. First, unlike standard multiplex epidemics where the *same* nodes participate in all layers (e.g., individuals on multiple social networks), our two layers contain *distinct node types* - agents and tools - producing a bipartite cross-layer coupling rather than a node-aligned multiplex. Second, the compartmental dynamics are *asymmetric*: agents follow a four-state SEIQ model while tools follow a two-state SIS model, reflecting their fundamentally different operational semantics. In contrast to the symmetric SIS–SIS multiplex studied by Sanz et al. (2014), our asymmetric SEIQ–SIS formulation introduces a non-trivial incubation delay in the agent layer that decouples R_0 from the incubation rate σ - a property we verify empirically in EG5. Third, these structural differences lead to a quadratic eigenvalue problem (Eq. 16) for the epidemic threshold that is not present in the cited single-layer or symmetric-multiplex MMCA formulations.

From a broader systems perspective, the cross-layer contagion mechanism we formalize is structurally analogous to cascading failures in interdependent infrastructure networks - a phenomenon studied extensively since the seminal work of Buldyrev et al. (2010) on coupled power grid and communication networks. In their model, failure of nodes in one infrastructure layer triggers failures in the dependent layer through coupling links, producing cascading collapses that are qualitatively absent in isolated networks. Our agent–tool multiplex exhibits the same structural property: the bipartite coupling matrix B plays the role of the inter-layer dependency links, and the cross-layer amplification factors we observe ($1.08\times$ – $1.50\times$) are the epidemic analog of the cascade amplification in interdependent infrastructure, though we note an important mechanistic difference: the Buldyrev model operates through percolation (permanent node removal) while our framework models endemic equilibria (reversible infection), making the analogy structural rather than mechanistic.

Comparison with prior analytical bounds

Our R_0 expression (Eq. 17) subsumes several known results as special cases. When $\beta_B = \beta_T = 0$ (pure agent network), we recover $R_0 = \lambda_A = \frac{\beta_A}{\gamma_A} \rho(A^T)$, which coincides with the classical NIMFA threshold (Van Mieghem et al. 2009) and the MMCA epidemic boundary derived by (a) Gómez et al. (2010). When $\beta_A = 0$ (pure bipartite), $R_0 = \sqrt{\lambda_B}$, matching the spectral radius of bipartite epidemic models surveyed by Wang et al. (2017). The full multiplex case introduces a cross-term $\sqrt{\lambda_A^2 + 4\lambda_B}$ that has no single-layer analog and quantifies the *synergistic risk amplification* of shared tool access. This structural result complements the inter-layer coupling analysis of Granell et al. (2013), who studied awareness-epidemic coupling on multiplexes but did not address bipartite cross-layer topologies with asymmetric compartmental dynamics.

Practical implications for DevSecOps

Assuming continuous convex returns on security investment, the following model-conditional insights emerge from the Marginal Rebalancing Theorem (Theorem 4). We emphasize that these are derived under the exponential efficiency decay assumption (Eqs. 22–23) and the illustrative parameter regime used throughout this study; their direct applicability to specific production deployments requires empirical calibration. Furthermore, treating semantic prompt injection filters as having continuous, exponential decay cost curves is highly idealized. Adversarial ML literature indicates that LLM guardrail bypasses are often “all-or-nothing”

discrete events rather than exhibiting smooth marginal diminishing returns.

- **Zero-Trust Tool Provisioning (Isolating Bipartite Hubs):** Our analysis demonstrates that shared tools with high degree centrality act as "super-spreaders," exponentially amplifying the dominant eigenvalue $\lambda_{\max}(BB^T)$. DevSecOps teams should strictly avoid global tool instances (e.g., a shared vector database). Instead, tools must be compartmentalized using strict IAM roles or instantiated on a per-agent basis.
- **Prioritizing Tool-Side Sandboxing:** Our optimization model indicates that when budget-constrained, isolating the shared state (tool sandboxing) provides a higher marginal reduction in steady-state prevalence (ρ_{∞}) than analyzing natural language agent-to-agent dialogues. DevSecOps should prioritize strong isolation mechanisms (e.g., Dockerized tool execution, strict gRPC boundaries) over computationally expensive semantic prompt firewalls.
- **Read-Write Asymmetry and Permission Scoping:** Contagion fundamentally requires a bidirectional path (Agent A writes payload to Tool; Agent B reads payload). We recommend enforcing strict read-write asymmetry. If an agent requires write access to a tool, that access should be scoped narrowly, and any agent consuming that data must treat the tool's output as untrusted, potentially malicious input.

These prescriptions should be interpreted within the scope of our model assumptions. In practice, agent-side defenses provide additional benefits (e.g., output quality control, compliance monitoring) not captured by the epidemic model. The allocation rule is most directly applicable when the primary objective is minimizing systemic infection prevalence. We further note that the inverse-efficiency law is a consequence of the exponential functional form; under alternative diminishing-returns models (e.g., logarithmic decay $\beta = \beta^{(0)}/(1 + kx)$), the optimal allocation becomes a constant fraction independent of the total budget C , qualitatively altering the rebalancing dynamics. The exponential model is adopted following standard security economics practice (Gordon and Loeb 2002), but practitioners should calibrate the efficiency function to their specific defense technologies.

Future directions

Several extensions merit investigation:

- *Temporal, directed, and adaptive topologies.* Extending the framework to time-varying adjacency

matrices $A(t)$ and $B(t)$ would capture the dynamic rewiring inherent in agent swarms where new tools are registered, agents are spawned, or communication patterns shift at runtime. Furthermore, decomposing the symmetric bipartite matrix B into directional matrices (B^{read} and B^{write}) would enable the modeling of asymmetric access controls (e.g., read-only vs. write-only tool permissions). This introduces a weighted, directed spectral problem that may require numerical rather than analytical treatment. Continuous-time formulations using coupled ODEs (Paré and Beck 2020) could complement our discrete-time MMCA in capturing dynamic permission changes over time.

- *Heterogeneous transmission rates.* Replacing scalar transmission rates with edge-specific parameters (e.g., β_{ij}^A and β_{ik}^B) would allow modeling of varying tool trust levels and agent privilege hierarchies.
- *Game-theoretic attacker models.* Introducing a strategic attacker who selects initial compromise targets to maximize cascading damage leads to a Stackelberg game formulation (Laszka et al. 2015). The defender's budget allocation (Theorem 3) becomes the follower's best response to the attacker's target selection.
- *Empirical calibration.* Calibrating $\beta_A, \beta_B, \beta_T$ from real-world red-teaming data on production multi-agent deployments (Ruan et al. 2023) would enable direct application of the framework to operational risk assessment.

Limitations

Several limitations warrant acknowledgment:

- *No empirical calibration.* All model parameters ($\beta_A, \beta_B, \beta_T, \sigma, \gamma_A, \gamma_T, \delta$) are illustrative values chosen to demonstrate the framework's analytical capabilities. The model has not been calibrated to or validated against actual prompt injection events in deployed multi-agent systems. Consequently, the reported prevalence values, amplification factors, and optimal allocation rules are model-conditional and should not be interpreted as predictions for specific real-world deployments without empirical calibration.
- *Static and symmetric topology.* We model fixed adjacency matrices A and symmetric access matrices B . In practice, agent swarm topologies are dynamic, with agents being added, removed, or rewired at runtime, and tool permissions often enforcing directional access controls (e.g., read-only).

- *Homogeneous transmission rates.* We assume uniform $\beta_A, \beta_B, \beta_T$ across all edges. Heterogeneous per-edge rates (reflecting varying tool trust levels or agent privilege hierarchies) would require a more granular parameterization.
- *Mean-field decoupling.* The R_0 derivation assumes an annealed mean-field approximation where A^T and BB^T share a dominant eigenvector. For highly heterogeneous multiplex topologies where this assumption fails, the analytical R_0 may deviate from the true spectral radius.
- *Rational attacker model.* We do not model adaptive attackers who strategically select initial compromise targets to maximize cascading damage.
- *Abstraction gap.* The mapping from actual prompt injection mechanics (context window poisoning, instruction hierarchy bypass, output format manipulation) to scalar transmission probabilities ($\beta_A, \beta_B, \beta_T$) is a modeling abstraction. The framework captures *topological co-exposure pathways* through shared tools, not the semantic details of how a specific prompt injection payload exploits a specific LLM's instruction-following behavior.

Conclusion

This paper has presented a theory-focused mathematical framework for modeling cross-layer security contagion dynamics in MCP-enabled multi-agent LLM swarms. Our coupled SEIQ–SIS MMCA on multiplex networks captures the cross-layer amplification mechanism whereby shared infrastructure tools create hidden contagion bridges between airgapped agents.

Three principal results emerge. First, the analytical approximation to the epidemic threshold, $R_0 = \frac{1}{2}(\lambda_A + \sqrt{\lambda_A^2 + 4\lambda_B})$, reveals a synergistic coupling between agent-layer and tool-layer spectral properties that subsumes classical single-layer thresholds as special cases (Remark 1) under the assumptions stated in Section Limitations. Second, the Marginal Rebalancing Theorem demonstrates that optimal security budget allocation follows an inverse-efficiency law - each additional unit of budget should flow preferentially to the less efficient defense layer. Third, ablation studies confirm that quarantine mechanisms are the single most sensitive control parameter, with their removal increasing R_0 from 28.85 to 1490.01 (about $52\times$ the full-model value).

Simulation-based validation across 8 experiment groups - including 30-run Monte Carlo alignment on canonical synthetic topologies, six structural ablations plus a topology comparison, and 20-run MMCA-vs-Monte-Carlo checks on three empirical agent-layer

datasets with synthetic tool layers - supports the MMCA's approximation accuracy relative to stochastic ground truth under the evaluated parameterizations. The sparsest empirical agent-layer network (CollegeMsg) exhibited the highest cross-layer amplification ($1.50\times$), suggesting that networks with weak intra-layer connectivity may be more vulnerable to tool-mediated contagion under the evaluated synthetic tool-layer constructions.

These results suggest that tool-side controls, including MCP sandboxing and HITL approval gates, can receive the larger modeled budget share in some evaluated regimes, particularly when tool-mediated coupling is strong. At the same time, the preferred allocation varies across parameter settings, architectures, and tool-layer constructions, so deployment decisions should be empirically calibrated rather than transferred directly from any single synthetic configuration. Future work will extend the framework along the directions outlined in Section Future directions, with particular emphasis on game-theoretic attacker modeling and empirical calibration from production deployments. As deployments scale from isolated agents to interconnected swarms, defense strategies must expand from isolated node patching to securing the multiplex infrastructure that bridges them.

Author contributions

Omitted in this anonymized submission version to preserve double-anonymous peer review. A full contribution statement will be restored in the non-anonymized version if required by the journal at a later stage of the editorial process.

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Data availability

The three real-world datasets used in EG8 (Email-EU, CollegeMsg, and ca-GrQc) are publicly available from the Stanford Large Network Dataset Collection (SNAP). All other reported results are generated from the simulation code and configuration settings described in the manuscript and accompanying scripts.

Code availability

All simulation code, experiment scripts, and network generation utilities are available through the anonymous review repository at <https://anonymous.4open.science/r/MMCA/>.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interest

The authors declare no competing interests.

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